

Graduation Project, Part-I
Computer Science Department
Jamoum University College
Umm Alqura University



جامعة أم القرى
الكلية الجامعية بالجموم
قسم علوم الحاسب



Glow Up Ai

Submitted by

Student Name	Student ID
Joud Adnan Alsheikh	444004461
Aseel Ahmad Alsubhi	444000725
Manar Omar Alluhibi	444000991
Sarah Omar Alrehaili	444004192
Lama Mohammed Almufarriji	44411757

Project Advisor: Dr. Hassan Althobaiti

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Abstract

In recent years, artificial intelligence (AI) and computer vision technologies have played an increasingly important role in personal skincare applications. Many people struggle to accurately identify their skin type and specific skin conditions, which can sometimes lead to the use of unsuitable products and ineffective skincare routines. To address this challenge, the **Glow Up AI** project presents a smart, mobile-based system that helps users receive personalized skincare recommendations using deep learning techniques.

Glow Up AI relies on convolutional neural networks (CNNs) to analyse facial images and determine skin type, skin tone, undertone, and common skin issues such as acne, pigmentation, and wrinkles. Based on the analysis results, the system generates a customized skincare routine and provides product recommendations, taking into account gender-specific biological differences to enhance accuracy. Additionally, the system offers cosmetic product and shade suggestions for female users and enables progress tracking through before-and-after photo comparisons.

The system is developed using the latest AI tools and is available as an easy-to-use mobile application supporting both Arabic and English. The project follows the Agile methodology to ensure flexibility, continuous improvement, and a user-centered design. Experimental evaluations and tests under various lighting and image conditions have shown that **Glow Up AI** can provide accurate and personalized skincare insights while maintaining usability and data security.

Overall, **Glow Up AI** bridges the gap between artificial intelligence and personal skincare in an accessible way by offering a precise, comprehensive, and easy-to-use solution that empowers users to make informed decisions about their skin health.

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Chapter 1

INTRODUCTION

1.1 Background and Motivation

In recent years, artificial intelligence (AI) and computer vision technologies have been increasingly integrated into personal care applications. Skincare, in particular, has become one of the fastest-growing fields where technology can contribute by providing accurate analyses and personalized recommendations. Many individuals struggle to determine their undertone and skin type, as well as various skin issues such as pigmentation, acne, and more. These difficulties often lead to the use of inappropriate products, causing greater problems, reduced effectiveness, and unnecessary expenditure of time and money.

The **Glow Up AI** project arises from the need to make skincare simpler and more science based. Unlike existing applications that provide general advice, **Glow Up AI** relies on image analysis and deep learning techniques to deliver recommendations that account for the physiological differences between men's and women's skin. This approach enhances user satisfaction, improves the accuracy of results, and helps users feel more confident and comfortable in making daily skincare decisions.

The project aims to assist users in easily selecting products suitable for their skin while avoiding mistakes that could lead to skin problems or wasted money.

1.2 Problem Statement

Many users face difficulties in accurately identifying their undertone, skin type, and skin conditions. Consequently, they often make inappropriate choices of skincare or cosmetic products, leading to issues such as acne, redness, or irritation. Current systems generally provide broad recommendations without considering individual differences or the biological variations between men's and women's skin.

Therefore, there is a clear need for an intelligent and comprehensive system capable of accurately analyzing facial skin, identifying its type, tone, and various conditions, and providing personalized skincare and cosmetic recommendations based on each user's unique characteristics.

1.3 Objectives

1. Accurate Skin Analysis:

- Identify each user's skin type (oily, dry, etc.).
- Detect visible skin issues such as acne, pigmentation, enlarged pores, and more.

2. Personalized Skincare Recommendations:

- Assist users in following a daily skincare routine suited to their skin type and condition.
- Suggest safe and effective skincare products and brands.

3. Cosmetic Product Suggestions (for Women):

Help users select cosmetic products appropriate for their skin type and undertone.

4. Progress Tracking:

Enable users to monitor their skin's progress by comparing "before and after" photos.

5. Awareness and Education:

Increase user awareness of skin health and promote proper skincare practices.

6. Integration of Technology and Beauty:

Combine AI-based skin analysis, personalized recommendations, and progress tracking into a single user-friendly application.

1.4 Project Scope and Limitations

The **Glow Up AI** project focuses on developing a mobile application that uses AI and computer vision to analyze facial images and provide personalized skincare insights. The scope includes facial skin analysis, personalized product and cosmetic recommendations, and skin progress tracking.

However, the project faces certain limitations, including:

- The accuracy of analysis is affected by image quality and lighting conditions during capture.
- The system currently focuses on facial skin only and does not cover other body areas.
- Recommendations are based on existing product databases, which may vary by region.
- Environmental factors such as humidity, diet, and lifestyle are not fully integrated into the current model, though they may be considered in future improvements.

Despite these limitations, **Glow Up AI** represents a significant step in integrating AI into personal skincare, helping users achieve healthier skin and make informed decisions when selecting products.

Chapter 2

Literature Review

2.1 Introduction

This chapter presents a comprehensive review of the literature relevant to AI-based skin analysis and personalized skincare systems. It highlights key studies, technologies, and methodologies related to skin type classification, personalized product recommendations, gender-based skin differences, and user experience in mobile dermatology applications. The purpose is to provide a clear understanding of the current research landscape, identify gaps, and assess the feasibility of developing an intelligent skincare system like **Glow Up AI**.

2.2 Review of Related Studies

Recent research has increasingly focused on integrating artificial intelligence and image processing into skin analysis, skincare recommendation systems, and mobile dermatology. This section summarizes the most relevant studies in five main areas: skin type classification, personalized skincare recommendations, skin tone and undertone detection, gender-based skin differences, and user experience considerations in dermatology applications.

2.2.1 AI in Skin Type Classification

2.2.1.1 Human Skin Type Classification Using Image Processing and Deep Learning Approaches

This study used CNNs and image processing methods to classify oily, dry, normal, and combination skin types based on 329 facial images. Data augmentation improved robustness, and accuracy exceeded 92%, confirming the effectiveness of deep learning in skin type detection under varying lighting conditions [1].

2.2.1.2 A Skin Type Classification Method Using Mobile Device-Based Deep Learning Model

This research proposed a lightweight, mobile-optimized deep learning model utilizing transfer learning to improve performance on smartphones. The model offered fast and efficient skin type classification but was influenced by camera quality and lighting variations [2].

2.2.1.3 Facial Skin Type Classification Based on Microscopic Images Using CNN

This study analyzed high-resolution microscopic skin images to detect pore density, oil patterns, and texture. It achieved high accuracy in controlled lab settings using CNNs but depended on specialized microscopic imaging devices, limiting everyday mobile use [3].

2.2.2 AI in Personalized Skincare Recommendations

2.2.2.1 AI-Powered Personalized Skincare Using Deep Learning Architectures

Miryabelli et al. introduced CNN, VGGNet, and DenseNet-based models trained on more than 10,000 facial images labeled by dermatologists. The system generated accurate skincare product recommendations, achieving 96.3% accuracy [4].

2.2.2.2 Adaptive AI in Cosmetics Integrating Genetic and Lifestyle Data

Hash et al. presented a model combining skin features, genetics, and lifestyle data for personalized cosmetic recommendations. The system reached 93% accuracy but faced challenges related to privacy and dataset diversity [5].

2.2.2.3 Intelligent Cosmetic Recommendation Systems Using Image Recognition and Ingredient Analysis

Lee et al. developed a hybrid model linking image-based skin analysis with ingredient-based product mapping. The model was effective in addressing dryness, pigmentation, and other conditions using ingredient correlation analysis [6].

2.2.3 Skin Tone and Undertone Analysis

2.2.3.1 AI-Driven Personalized Skincare for Tone and Condition Detection

This study used CNNs to detect skin tone, pigmentation, acne, and dryness. It demonstrated strong adaptability to mobile platforms and consumer applications while requiring high-quality images for optimal performance [7].

2.2.3.2 Undertone Classification Using Wrist-Vein Images and CNN

This research classified undertones (warm, cool, neutral) using CNNs applied to wrist vein images. It achieved 99.5% accuracy with well-controlled image capture conditions [8].

2.2.3.3 Skin Tone and Undertone Detection Using LAB and HSV Color Spaces

This study used CNNs in combination with LAB and HSV color transformations to detect skin tone and undertone. It achieved 80% accuracy under varied lighting with preprocessing techniques [9].

2.2.4 Gender-Specific Skin Analysis

2.2.4.1 Male vs. Female Skin: Structural Differences

Rahrovan et al. highlighted structural skin differences between males and females, including collagen density, oil production, and aging patterns. These differences influence skincare needs and treatment outcomes [10].

2.2.4.2 Gender-Based Variations in Post-Surgical Skin Healing

This study examined post-surgical healing differences and found that females generally exhibit faster recovery and tissue remodeling than males due to biological factors [11].

2.2.4.3 Sex Differences in Skin Immunity via the Androgen–ILC2–DC Pathway

Chi et al. identified a molecular pathway showing how androgen signaling affects immune response in male skin, establishing foundational biological differences in skin immunity [12].

2.2.5 User Experience and Mobile AI Dermatology

2.2.5.1 Enhancing User Trust through Transparent AI Interfaces

Kwok et al. found that user trust improves when AI systems provide simple, clear visual explanations of their diagnostic results [13].

2.2.5.2 Personalization and Multilingual Accessibility in AI Dermatology

This study emphasized including multilingual support and cultural adaptability to improve accessibility and user satisfaction in AI dermatology tools [14].

2.2.5.3 Balancing Accuracy and Inclusivity in Mobile AI Skin Analysis Tools

Sajid et al. highlighted performance gaps in dermatology apps, especially for darker skin tones, underscoring the need for fairness and dataset diversity [15].

Final Comparison

After analyzing previous studies across the five research areas, several key observations can be drawn:

- **Skin Type Classification:**
Deep learning models demonstrate their effectiveness in achieving high accuracy rates. However, their efficiency is affected by several factors, most notably lighting conditions, image quality, and the type of device used for capturing the image.
- **Providing Personalized Skincare Recommendations:**
The proposed systems range from models that rely solely on image analysis to those that combine image data with genetic factors and lifestyle. However, most of these systems lack the ability to operate in real-time on smartphone applications and do not offer a sufficient level of comprehensive personalization.
- **Skin Tone and Undertone Determination:**
Studies have shown that methods based on wrist images achieve higher accuracy, while methods based on facial images are easier to use in practice. Variations in lighting remain a major challenge affecting the accuracy of the results.
- **Studies on Sex Differences in Skin:**
Biological research confirms clear differences between male and female skin in terms of structure, immune response, and recovery mechanisms. However, most current AI systems do not take these differences into account.
- **User Experience Research:**
Studies indicate that factors such as trust, transparency, and inclusivity, along with support for diverse skin tones and languages, are essential for ensuring user acceptance and system adoption in real-world situations.

Glow Up AI aims to address these gaps by integrating gender analysis, accurate skin tone and undertone detection techniques, real-time CNN models, diverse training data, a clear user interface, and multilingual support within a single smartphone-optimized system.

2.3 Research Gap and Justification

Certain gaps have been pointed out in the existing literature, which include:

1. **Limited Generalization:** The model performs well when tested under controlled conditions. It may have difficulties when the imaging environment is not stable.

2. Lack of overall personalization: The systems make recommendations based on limited data without taking into consideration environmental factors, lifestyle, and preferences of the users, which affects the recommendations' effectiveness.

3. Insufficient consideration of diversity in terms of gender and skin tones: In most models, there is a lack of consideration of gender and skin tone variations.

4. Practical limitations: Certain schemes are based on special image analysis systems and thus are inadmissible for mobile communication.

5. Weak Adaptive Learning: These systems are unable to adapt to continuous changes because their suggestions are static and do not adapt effectively to the variations that occur in skin conditions.

The **Glow Up AI** project seeks to fill these gaps in the research space by combining lifestyle trends, deep learning, gender modeling, and adaptive personalization, and in the process, offer a mobile-friendly user experience

2.4 Feasibility Study

Technical Feasibility

The system can be developed using established frameworks such as TensorFlow and PyTorch, leveraging transfer learning and TensorFlow Lite/CoreML for mobile deployment. These tools enable real-time image processing while maintaining scalability and efficiency (Lee et al.; Saiwaeo et al.).

Operational Feasibility

Agile development methodology supports iterative testing and improvement. The system requires data collection pipelines, model integration, and cloud infrastructure for synchronization and updates, ensuring both accuracy and usability (Guo and Cheng).

Practical Feasibility

Challenges such as lighting variations, camera differences, and skin diversity can be mitigated through dataset augmentation, color calibration, and adaptive preprocessing. Data privacy will be ensured through encryption and authenticated access (Kim et al.; Lee et al.).

2.5 Summary

This chapter reviewed existing research on AI-based skin analysis and personalized skincare systems. While deep learning models have shown strong accuracy in skin type classification and product recommendation, their real-world performance is often limited by lighting conditions, camera variations, and insufficient dataset diversity. Studies also highlight the importance of considering skin tone, undertone, and gender-specific biological differences,

which many current systems overlook. Additionally, user trust and accessibility depend on transparent and inclusive interface design. These findings justify the development of **Glow Up AI**, which aims to integrate image analysis, lifestyle data, gender-aware modeling, and user-centered design into a comprehensive mobile skincare solution.

Chapter 3

Methodology

3.1 Research Methodology

The project is using the Agile development methodology due to its suitability in software development projects that require constant improvement and flexibility. The **Glow Up AI** system also demands frequent testing, upgrading, and changes according to user reactions, thus the suitability of the Agile development methodology.

Agile emphasizes collaboration, adaptability, and building the system incrementally. This is particularly helpful when building an AI skincare application because there may be aspects of the application, for instance, image analysis, accuracy, or user interface design, that would benefit from revisions following testing and feedback.

The project is implemented in short cycles, referred to as “sprints.” Each sprint involves reviewing, testing, and developing or improving the system. This enables monitoring of progress, correcting errors, and increasing system performance and usability over time.

Whereas in the Waterfall model, the sequence of phases is rigidly followed, Agile permits changes at various stages of the development process. This is significant for AI and user-centric apps because in such projects, the requirements keep altering as the project progresses further.

In particular, the use of an Agile development methodology will provide an efficient and organized approach to the development of **Glow Up AI**. This will be achieved by improved accuracy, as indicated by continuous feedback.

3.2 Tools and Technologies / Data Collection Methods

The **Glow Up AI** project integrates various current tools, technology, and programming languages to implement the system and test out the ideas. Most of the work has been done using the Python programming language since it can interact effectively with machine learning technology such as TensorFlow, Keras, and OpenCV, which helps the system evaluate the images and implement the model. Such technology helps the system make use of CNNs to evaluate various skin conditions, skin tone, and skin textures.

In the case of the mobile app, Flutter was employed to design the interface. This interface can function on any iOS platform. Then, the backend processes the functionality associated with the login feature using Firebase. This ensures the data is secure. Additionally, Google Cloud

Vision API, along with libraries such as NumPy or Pandas, assists in processing the large number of images.

As far as the data is concerned, we have applied a research-based technique to make the data unbiased, diverse, and reliable. Skin images are obtained either from the public datasets available for dermatology, confirmed image sources, or contributions by users. The dataset comprises a variety of skin types, skin colors, and conditions, making the model more accurate and less biased. The images undergo processes like resizing, normalization, and transformation processes that involve rotation, brightness, and noise elimination.

3.3 PROJECT TIMELINE

Phase No.	Phase Name	Duration									
		1-2	3-4	5-6	7-8	9-10	11-12	13-14	15-16	17-18	19-20
1	Planning & Requirement Analysis										
2	Literature Review & Data Collection										
3	System Design										
4	Implementation & Development										
5	Testing & Evaluation										
6	Documentation & Reporting										

TABLE 1- PROJECT TIMELINE

The project timeline is presented to illustrate the sequence, duration, and progression of the **Glow Up AI** development phases over a 20-week period. Each phase is allocated a specific time range to ensure systematic execution and effective project management.

The project begins with **Planning & Requirement Analysis (Weeks 1–2)**, focusing on defining the problem, objectives, scope, and system requirements. This is followed by

Literature Review & Data Collection (Weeks 3–4), during which related studies are reviewed, research gaps are identified, and relevant datasets are collected and evaluated.

The **System Design phase (Weeks 5–8)** establishes the architectural foundation of the system, including data flow modeling, use case design, and database structure. Subsequently, **Implementation & Development (Weeks 9–14)** represents the core development stage, where the AI model is trained, image preprocessing is implemented, and the mobile application and backend components are developed and integrated.

After development, **Testing & Evaluation (Weeks 15–18)** is conducted to validate system functionality, assess AI model accuracy, and evaluate performance and usability under different conditions. Finally, **Documentation & Reporting (Weeks 19–20)** focuses on analyzing results, preparing the final report, and completing all required documentation for submission.

3.4 RISK MANAGEMENT

Planning in risk management played a significant role in the success experienced in the **Glow Up AI** project, as it prevents risks and prepares one against any challenges that might come along the way. This section will highlight the risks involved in the project as follows:

Risk Type	Risk Impact	Risk Level	Mitigation Strategy
Technical Risk	The accuracy of the analysis may decrease when using low-resolution or poorly lit images.	High	Expand the database to include various skin types, apply image quality enhancement techniques, and automatically adjust lighting.
Data Risk	There is a possibility of insufficient representation of different skin types in the database or potential bias.	High	Applying techniques to balance rows, test data fairness, and compile a diverse database covering multiple skin types and colors.
Security and Privacy Risk	Storing and processing user images and personal data may raise privacy concerns.	High	Clearly communicate privacy policies to users, implement encryption during data transmission and storage, and apply strong security protocols.
Schedule Risk	Task delays or weak coordination within the team may affect project progress.	Medium	Follow the Agile methodology by dividing tasks into short sprints and holding weekly progress reviews to ensure timely completion.
User Acceptance Risk	Some users may not trust AI-generated results or may find the interface difficult to use.	Medium	Design a simple and intuitive interface, explain how the analysis works, and collect user feedback for continuous improvement.
Maintenance and Update Risk	AI models may require periodic updates to improve performance or maintain compatibility with newer devices.	Medium	Maintain clear documentation, use a flexible architecture that supports future updates, and plan for scheduled post launch maintenance.

TABLE 2 - RISK MANAGEMENT

It helps to develop a risk management plan that is well-defined and reduces the likelihood of developing issues with **Glow Up AI**. This will involve continuous monitoring and proactive planning by team members to help maintain project efficiency and achieve objectives within the defined time and resource limits.

3.5 Quality Assurance Plan / Validation and Reliability

Glow Up AI models will also go through a number of processes to determine the reliability and accuracy of the results obtained. A variety of test processes will be used at the different stages of the development process. The deep neural networks will also be tested using a test set to test the reliability of the classification of the types, stage, and conditions.

Testing the performance of the system using different shooting conditions such as lighting, image quality, and angles at which the image is shot. This ensures the system works properly in any setting that the application might be utilized in. User-friendly nature will also be evaluated in the study by assessing the clarity of the output as well as the flow at which the system's interface is interacted with.

These combined processes help improve the accuracy, robustness, and reliability of **Glow Up AI** even before it is deployed in the real world since they ensure it is capable of delivering accurate results.

3.6 Summary

This chapter discusses how this project is constructed, which utilizes an Agile development approach that enables flexible and iterative construction. The chapter details various tools used such as Python, TensorFlow, Flutter, and Firebase, as well as strategies used in handling and processing gathered information in order to make it just and accurate. The project timelines, risk management, and validation are also defined in this chapter.

Chapter 4

Project Requirements & Analysis

4.1 Stakeholder Identification

Glow Up AI involves several key groups who contribute to or benefit from the system:

- **End Users:**
Individuals using the app to analyze their skin and receive personalized recommendations.
- **Female Users:**
Women who receive additional cosmetic guidance like foundation shade matching.
- **Development Team:**
Responsible for designing, building, and testing the app.
- **Cloud Service Providers:**
Manage secure data storage and backend operations.
- **University:**
Oversees the project as part of academic requirements.

Assumptions

- Users will upload clear images under good lighting.
- Internet access is available for app functionality.
- The dataset includes diverse skin types and tones.

Constraints

- Privacy rules must be followed.
- Accuracy depends on image quality and device capabilities.

4.2 Methods of Requirements Gathering / Data Needs and Sources

To ensure that the requirements of the **Glow Up AI** system align with both user expectations and technical feasibility, multiple methods were used to collect, analyse, and validate requirements. The process combined both qualitative and quantitative approaches to obtain a clear understanding of user needs, functional expectations, and data specifications necessary for effective AI-based skincare analysis.

1. Interviews

We interviewed people around us, including family members, friends, and colleagues, to gather general opinions on features that might help identify common skin problems. The feedback we collected helped us formulate the system's core criteria, such as classifying skin type, identifying common conditions like acne, pigmentation, and dryness, and developing simple guidelines to ensure the safety of product recommendations for users.

2. Market and Competitor Analysis

Existing skincare apps, including tools like Sephora's "Smart Skin Scan," were analysed to identify strengths, weaknesses, and gaps in current solutions. This benchmarking process provided valuable input for designing competitive advantages in **Glow Up AI**, such as gender-specific analysis, multilingual interface support, progress tracking features, and a complete focus on skincare using cutting-edge technologies to deliver personalized guidance and safe product recommendations.

3. Data Sources and Datasets

For this project, we used publicly available skin image datasets as well as some facial images we collected ourselves to train and test the system. The datasets have labeled pictures showing different skin types, tones, and conditions. We tried to make sure the data was fair and inclusive by including people of different genders and photos taken under various lighting conditions.

4. Ethical and Privacy Considerations

All data collection and handling processes adhere to strict privacy and ethical standards. User data and images are anonymized, encrypted, and used solely for research and system improvement purposes. Participants are informed about data usage and consent is obtained prior to data collection.

By combining user-centered feedback and data-driven research, the requirements for **Glow Up AI** were defined comprehensively. This multi-source approach ensures that the system is technically robust, user-friendly and providing accurate and trustworthy skincare recommendations.

4.3 Functional Requirements

1. User registration and authentication

The system allows users to create personal accounts and log in using their email or phone number, with the ability to save and update personal data when needed.

2. Photo Capture

Users can take or upload facial photos. The system automatically detects the face, adjusts lighting, and enhances image quality before analysis.

3. AI skin analysis

The system uses deep learning algorithms (CNN) to determine skin type and color, identify undertones, and detect skin problems such as acne, pigmentation, or wrinkles.

4. Skin care routine recommendations

Based on the analysis results, the system provides a skin care routine tailored to the skin type and recommendations for suitable products, taking into account gender differences to ensure accuracy.

5. Suggest beauty products (for female users)

The system provides recommendations for beauty products such as foundation and shades that match the user's skin tone and undertone.

6. Comparison and progress tracking

The system allows users to compare previous and recent analysis results through images that show the improvement in skin condition.

7. Ease of access and language support

The system supports both Arabic and English and offers a user-friendly and clear interface for all users.

8. Product Rating and Feedback

- Users are allowed to rate products by giving a star rating and adding a short comment. They can also edit or delete their rating at any time.

- These ratings are used to improve future recommendations and filter out products that are not suitable for the user.

4.4 Non-Functional Requirements

The non-functional requirements of the **Glow Up AI** system describe the quality, performance, and operational requirements that ensure the system operates efficiently, securely, and stably for users.

1. Performance Requirements

- The system must be able to process and analyse facial images, whether captured directly (Capture) or uploaded (Upload), within a reasonable response time to provide a smooth user experience.
- The AI model should maintain reliable accuracy in identifying skin type, tone, and common conditions, regardless of variations in lighting or image quality.

2. Usability Requirements

- The application interface should be simple and intuitive, ensuring a clear user experience for both technical and non-technical users.
- The analysis process should follow a few clear steps:
(Capture or Upload image → Analysis → Display of results and recommendations → Rating and feedback submission).

3. Reliability Requirements

- The system must maintain stable performance under normal usage and typical mobile internet speeds.
- The application should handle potential issues such as blurry or low-quality images, providing guidance to help the user retake or re-upload correctly.

4. Security and Privacy Requirements

- All user data, including images, ratings, and feedback, must be encrypted during storage and transmission.
- Access to personal data should be restricted to trusted accounts, with users able to delete their data at any time.
- The system must adhere to ethical principles and data protection standards to ensure confidentiality and build user trust.

5. Maintainability Requirements

- The system should be built in a modular way, allowing model improvements or new features to be added without extensive code changes.
- Model updates should be possible without causing a complete service outage.

6. Scalability Requirements

- The back-end and cloud infrastructure must be able to scale to support a growing number of users, captured and uploaded images, and feedback data as the application expands.
- The system should efficiently handle the increase in data generated from ratings and feedback.

4.5 Constraints

The **Glow Up AI** project faces several constraints that may affect its development and performance:

1. Technical Constraints:

- The accuracy of skin analysis depends on image quality, lighting conditions, and camera specifications.
- The application relies on specific development tools and frameworks (Python, TensorFlow, Keras, OpenCV, Flutter, Firebase), which may limit flexibility in adopting alternative technologies.

2. Data Constraints:

- Recommendations are based on available datasets and product databases, which may vary regionally and might not cover all skin types or conditions comprehensively.
- Ensuring diversity and fairness in the dataset is challenging, and model performance may vary for underrepresented skin tones.
- All collected user data must comply with strict privacy and ethical standards, including anonymization and encryption.

3. Project and Environmental Constraints:

- The development timeline and team resources impose limits on feature implementation and testing.
- Environmental factors such as diet, lifestyle, or humidity are not fully integrated into the current analysis model.
- Continuous updates and model retraining are required to maintain accuracy and compatibility with new devices.

These constraints have been carefully considered to ensure that **Glow Up AI** delivers reliable, accurate, and user-friendly skincare recommendations within the defined technical, data, and operational boundaries.

4.6 Summary

This chapter defines the system's stakeholders, functional and non-functional requirements, and project constraints. It explains how requirements will be gathered through interviews, market analysis, and dataset research. The chapter also details features like AI skin analysis, personalized recommendations, and secure data handling, while addressing limitations such as image quality and dataset diversity.

Chapter 5

SYSTEM DESIGN

5.1 Overview of Design Approach

The design of the **Glow Up AI** system is built on an approach to balance high accuracy in skin analysis with usability. The system provides a seamless and smooth user experience, powered with advanced AI technologies and a structure that can be extended in the future. The design is based on the following principles:

1. User-Centered Design

The interfaces are clear and simple, with support for both Arabic and English to make access easy for a wide range of users. The goal is to decrease complexity and increase confidence in the results of the analysis.

2. Modular Design

The system is divided into independent modules, including a user management module, image processing module, an artificial intelligence model, and a module for recommendations. This approach allows for easy development and updates of each component without affecting the rest of the system, in line with the Agile methodology adopted.

3. Intelligent Automation

The application is based on automated processes, such as lighting optimization, face detection, and photo quality correction prior to analysis. The AI model identifies the skin type, base tone color, and skin problems, then convert the results into personalized recommendations tailored specifically to each user.

4. Scalability and Future Development

The cloud-based structure enables secure storage of data and the ability to deal with a larger number of users without affecting performance. The system is also designed to support future expansions, such as the integration of environmental factor analysis or the expansion of the product database in general, this design combines the power of artificial intelligence with a simple and convenient user experience, with the aim to provide a flexible, intelligent, and easy-to-use skin analysis system.

5.2 Data Flow Analysis

This diagram shows how data flows through the system, from input and process to output, to show process sequence and flow of information.

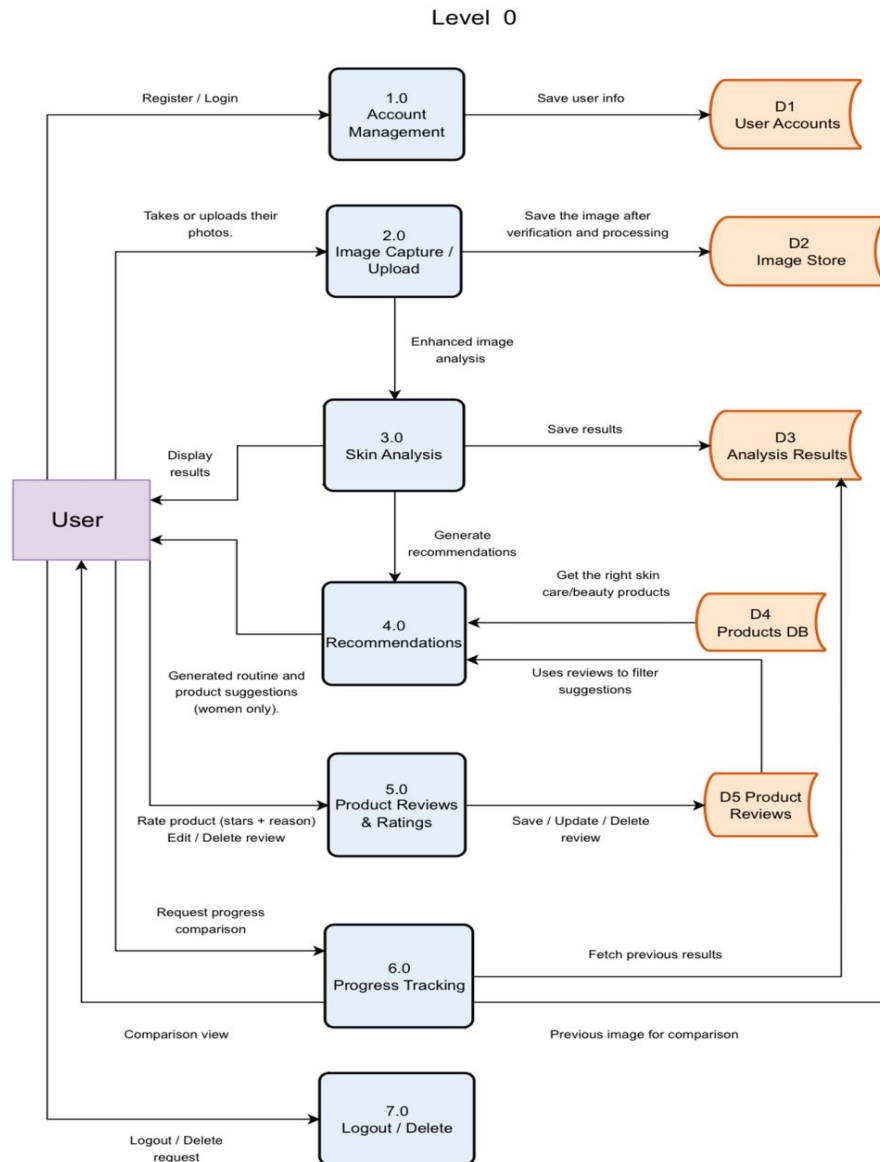
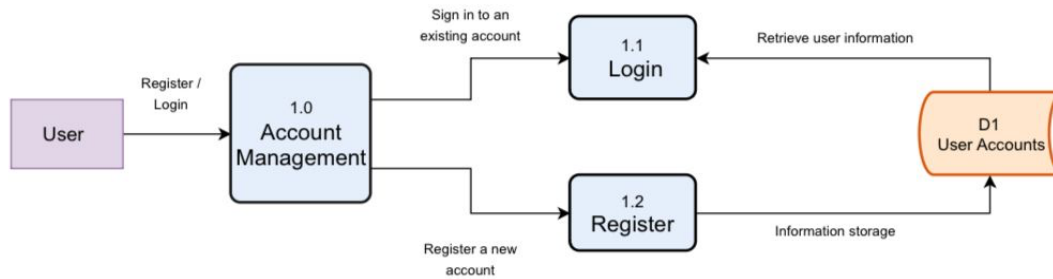


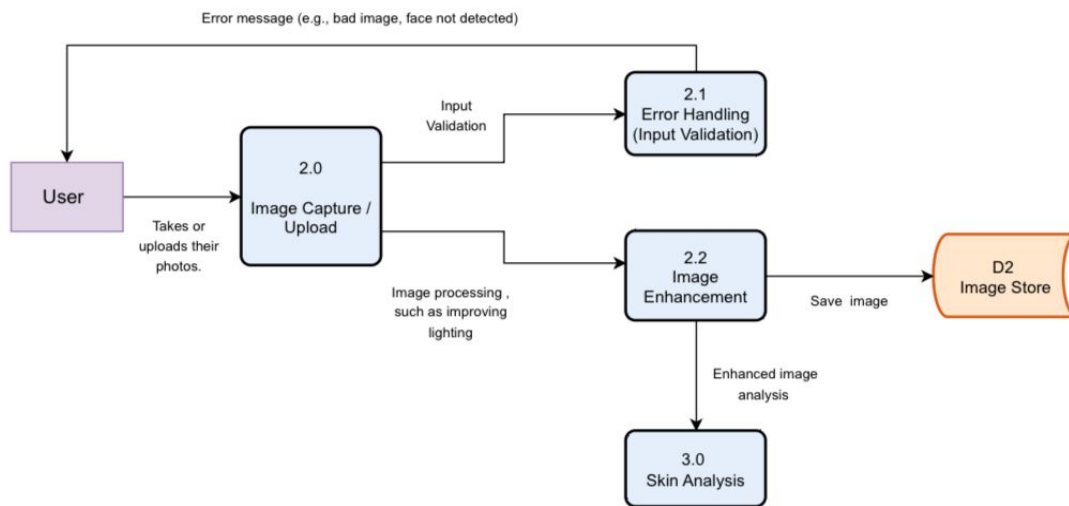
FIGURE 1 - DATA FLOW ANALYSIS LEVEL 0

Level 1

Process 1



Process 2



Process 3

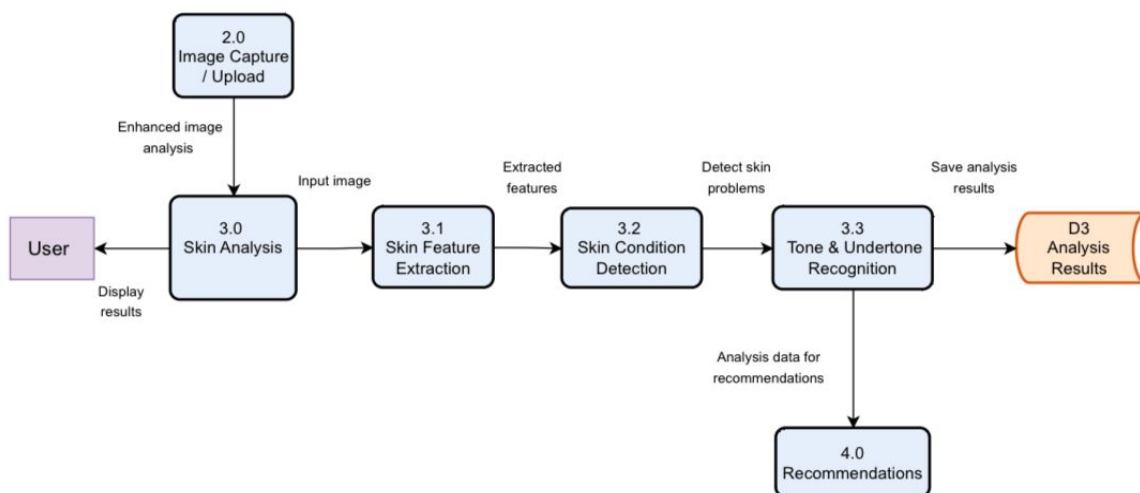
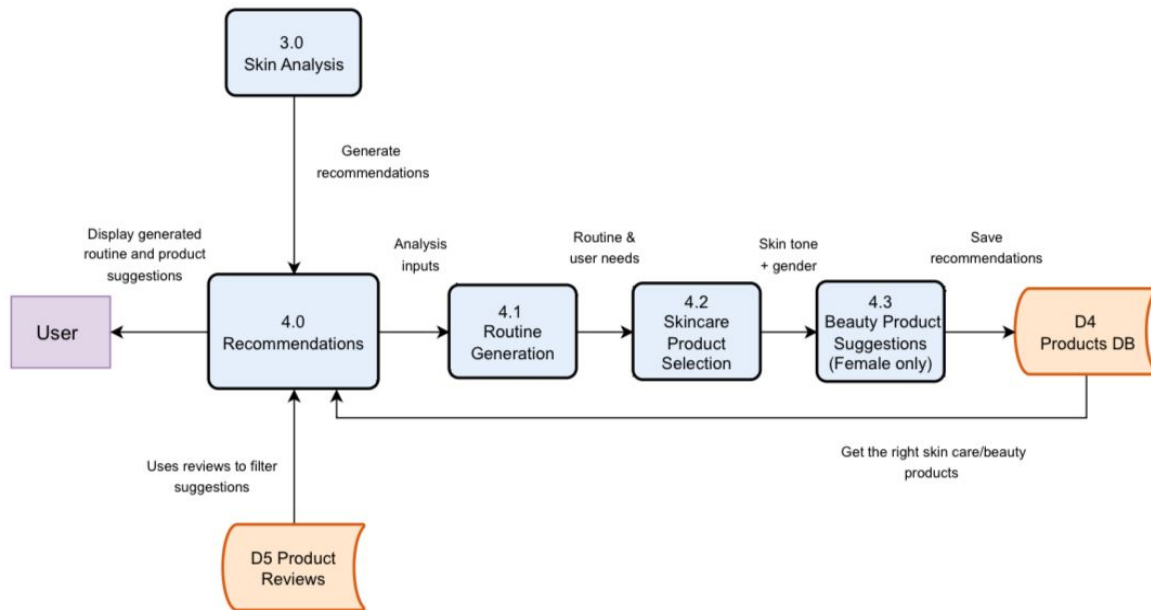
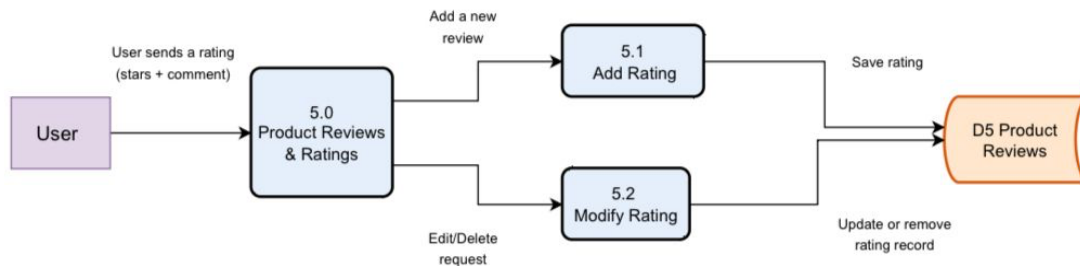


FIGURE 2 - DATA FLOW ANALYSIS LEVEL 1.1

Process 4



Process 5



Process 6

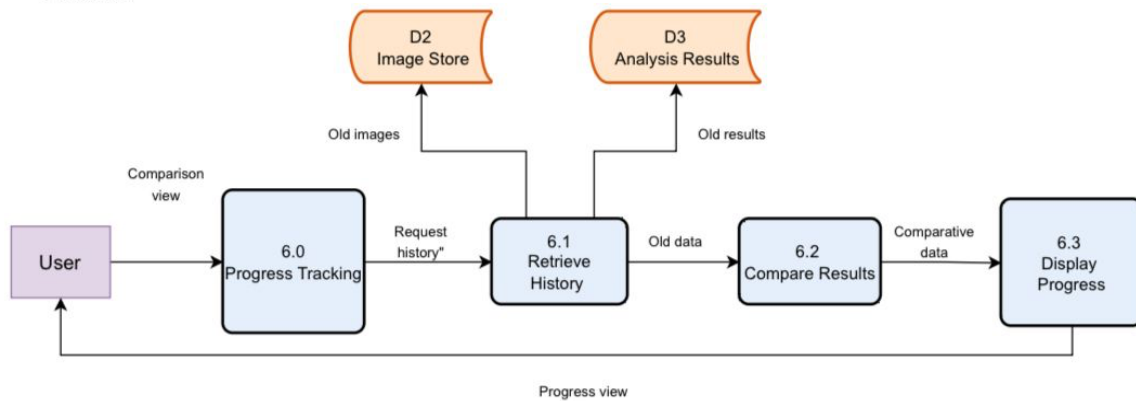


FIGURE 3 - DATA FLOW ANALYSIS LEVEL 1.2

5.3 System Architecture

The **Glow Up AI** system is designed with a five-layer architecture to ensure clarity, scalability, and flexibility. Users interact through mobile or web applications within the delivery layer. The application layer manages interface logic and workflow processing. Core backend functions, including AI analysis and API processing, operate in the services layer. All persisted data, images, and product information stored securely in the resource layer.

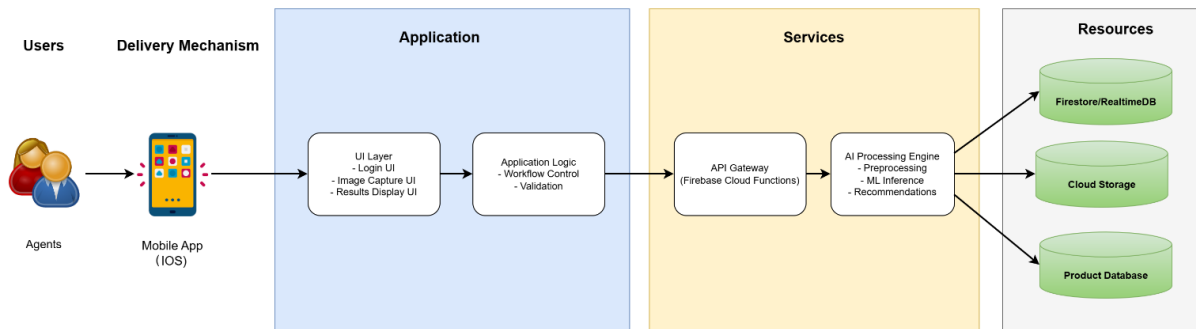


FIGURE 4 - SYSTEM ARCHITECTURE

5.4 Use Case Analysis

The system consists of two main components that work together:

1. The AI Model, which is responsible for analyzing images and extracting skin results, based on computer vision technologies (CNN).
2. The application, which represents the interaction and display interface for users.

5.4.1 Main Use Cases:

1. Account Registration/Login

The user creates a new account or logs into an existing one to enable saving data, results, and previous analyses.

2. Facial Image Capture

The user takes a picture, and the system automatically enhances the lighting and quality before the smart model analyzes it.

3. Image Enhancement

The system automatically enhances image quality using image processing algorithms (face detection, noise reduction, lighting calibration).

This process is performed by the smart model and the system's processing unit.

4. Skin Analysis

The system uses a CNN model to identify skin type and its problems (such as dark circles, acne, etc.) and also recognizes skin tone and undertones.

5. Skincare Recommendations

After analysis, the system provides a personalized routine based on skin type and condition. It suggests a range of suitable skincare products tailored to each skin type and its specific needs.

6. Beauty Product Suggestions (for Women)

The system suggests suitable beauty products, such as foundation and concealer, based on skin tone and type.

7. Results Display

The application displays the final results of the skin analysis clearly and systematically.

8. Saving Results and Progress Tracking

The system allows users to save analysis results and track skin improvement by comparing new results with previous ones.

9. Product Reviews & Ratings

Users can delete personal information or saved photos.

5.4.2 Actors:

The user is the only actor interacting with the system.

They are the person who takes the photo for skin analysis, receives recommendations, and can save the results and track their progress over time. The user can be male or female, with additional features for females such as beauty product suggestions.

5.5 Entity-Relationship (ER) Analysis

This section explains how **Glow Up AI** application data is organized using an entity-relationship (ER) diagram. The database is designed to include six main tables that ensure all data necessary for the application to run efficiently is stored.

The main tables are:

1. **User Table:** This table stores basic user data, such as name, email, gender, and registration date.
2. **Skin Analysis Session Table:** This table records the results of each skin examination, including skin type, base color, conditions detected, and the degree of confidence in the analysis results. Each analysis session is linked to a single user.
3. **Skin Condition Table:** Contains a list of possible skin conditions such as acne or dryness. A single analysis session can reveal multiple conditions, and a single condition can appear in multiple analysis sessions.
4. **Product Recommendation Table:** Stores product suggestions resulting from each skin examination, such as suitable cleansers or moisturizers.
5. **Routine Table:** Provides users with daily skin care plans based on the results of the examination.
6. **Progress Track Table:** Allows users to compare the results of previous and subsequent examinations to monitor improvements in skin condition.

Entity Relationships

1. A single user can have multiple skin analysis sessions conducted at different times.
2. One skin analysis session can identify several skin conditions, and the same condition may appear across different sessions.
3. Each skin analysis session generates a set of product recommendations tailored to the analysis results.
4. One personalized skincare routine is created for each skin analysis session based on its outcomes.
5. The ProgressTrack entity links two different analysis sessions for the same user to enable result comparison over time.

This structure ensures efficient data organization, simplifies access to user records, and supports the core features of the application, including recommendations and progress tracking.

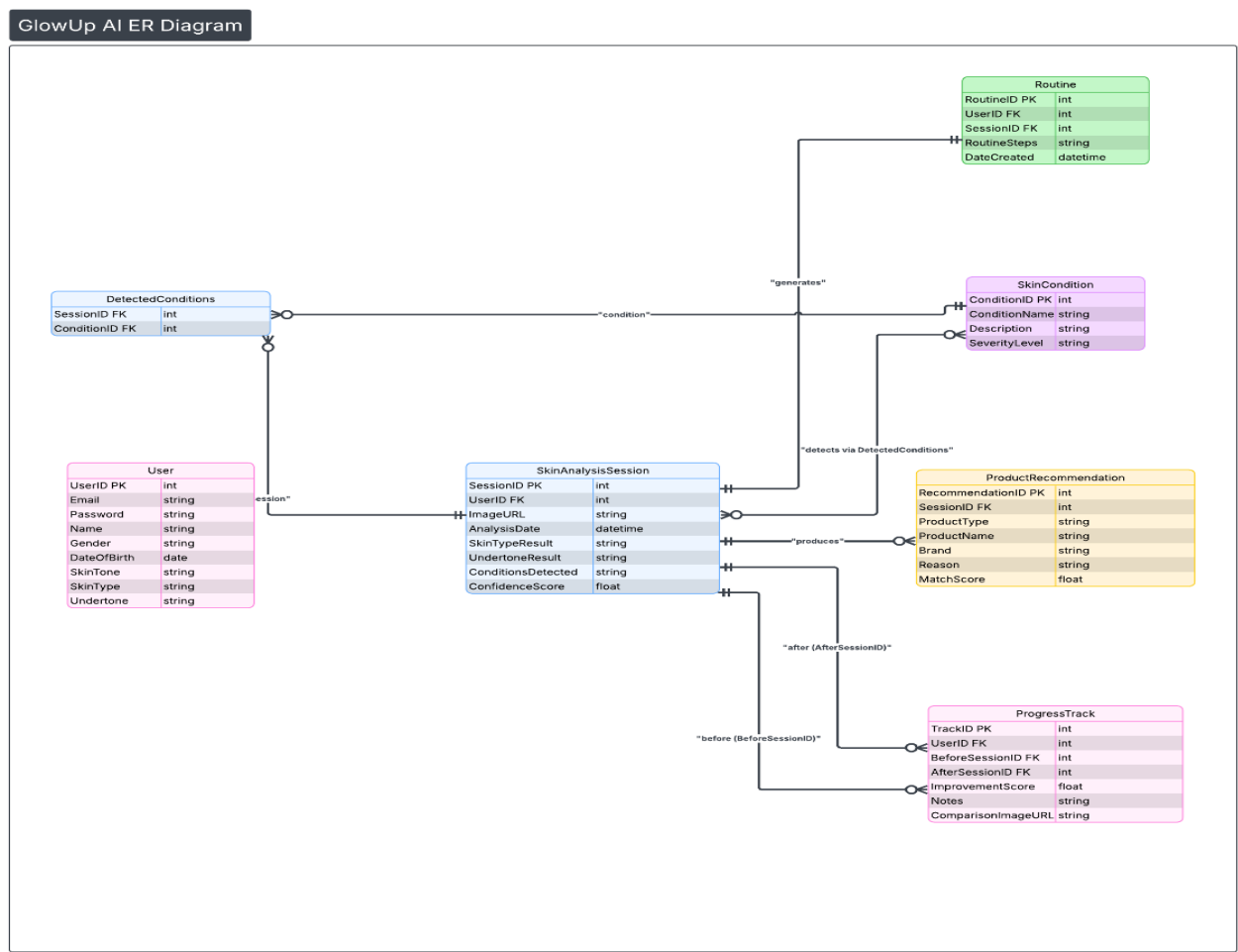


FIGURE 5:ENTITY RELATIONSHIP

5.6 System Use Case Diagram

The system includes a single primary actor, the User, representing both male and female users. Gender-specific features, such as beauty product recommendations for female users, are handled through optional extended use cases rather than defining separate actors.



FIGURE 6 - SYSTEM USE CASE DIAGRAM

AI Processing Use Case Diagram

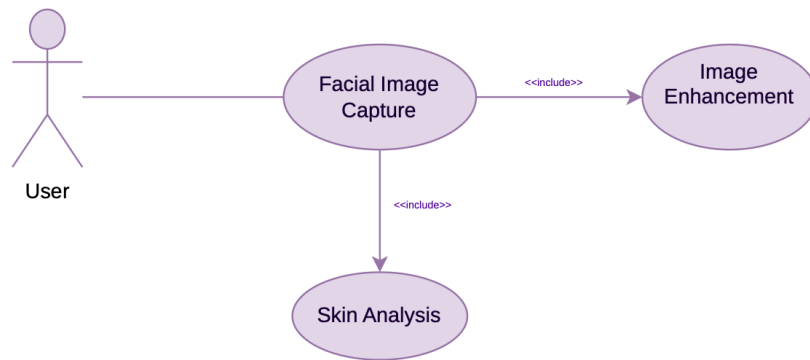


Figure 6.1: AI Processing Use Case Diagram

Recommendation Use Case Diagram



Figure 6.2: Recommendation Use Case Diagram

Results Management Use Case Diagram

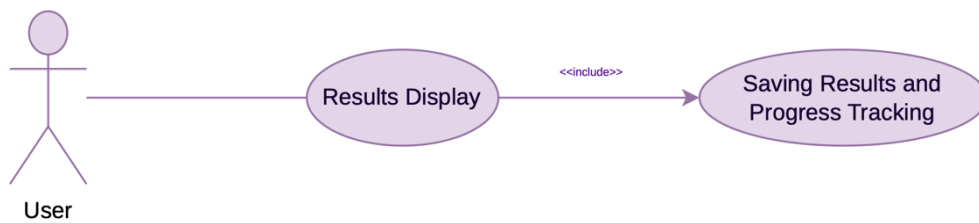


Figure 6.3: Results Management Use Case Diagram

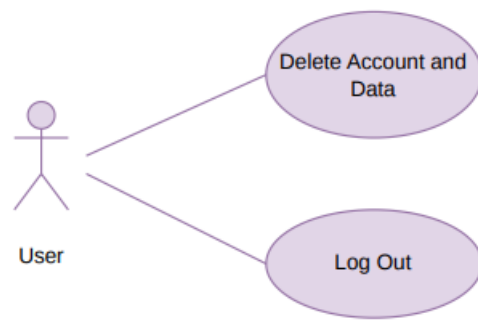
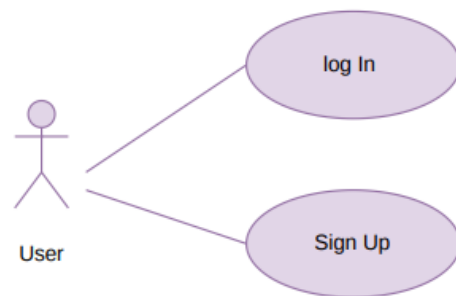


Figure 6.4: Account Management Use Case Diagram



5.7 System Use Case Description

1. Account Registration/Login

Element	Description
Use Case Name	Account Registration/Login
Goal	Allow users to create a personal account or log into an existing one to enable saving data and tracking progress.
Actor	The User
Preconditions	The user must have internet access; the application must be installed and open.
Basic Flow	1. The system displays the login/signup screen. 2. The user enters their credentials (email/password or phone number). 3. The system authenticates the user via Firebase. 4. If credentials are correct, the user is logged in and directed to the home screen.
Alternative/Exception Flow	A1: Incorrect Credentials: The system displays an error message and prompts the user to try again. A2: New User: If the user selects signup, the system creates a new user profile on the backend.
Postconditions	The user is authenticated; personal data and analysis history are loaded.

TABLE 3 - ACCOUNT REGISTRATION/LOGIN

2. Facial Image Capture

Element	Description
Use Case Name	Facial Image Capture
Goal	Enable the user to capture or upload an image of their face for skin analysis.
Actor	The User
Preconditions	The user is logged in; the app has camera/photo library access permission.
Basic Flow	1. The user selects the "New Analysis" option. 2. The system guides the user to capture an image under good lighting conditions. 3. The user captures the photo or selects one from their gallery. 4. The system automatically detects the face within the image. 5. The image is passed to the "Image Enhancement" phase.
Alternative/Exception Flow	A1: Poor Lighting: The system displays a warning prompting the user to improve lighting. A2: Unclear Image: The system rejects the image and prompts for a clearer capture.
Postconditions	A facial image of acceptable quality is obtained and queued for processing.

TABLE 4 - FACIAL IMAGE CAPTURE

3. Image Enhancement

Element	Description
Use Case Name	Image Enhancement
Goal	Automatically enhance the quality of the captured image to maximize the accuracy of the AI analysis model.
Actor	System
Preconditions	A facial image has been captured or uploaded.
Basic Flow	1. The system receives the captured image. 2. Image processing algorithms are applied to adjust lighting, reduce noise, and perform resizing. 3. Color calibration techniques are applied to normalize the image colors. 4. The enhanced image is prepared as input for the Convolutional Neural Network (CNN) model.
Alternative/Exception Flow	A1: Insufficient Enhancement: If the enhancement fails to meet quality standards, the system returns a prompt requesting the user capture a better source image.
Postconditions	The image is optimized for AI processing, mitigating external variables like poor lighting.

TABLE 5 - IMAGE ENHANCEMENT

4. AI Skin Analysis

Element	Description
Use Case Name	AI Skin Analysis
Goal	Use the deep learning algorithm to accurately determine the user's skin type, tone, and detect specific skin problems.
Actor	System
Preconditions	The image enhancement process is complete; the optimized image is ready.
Basic Flow	1. The AI Model Service receives the enhanced facial image. 2. The CNN model determines the skin type (oily, dry, combination, or sensitive). 3. The model detects visible skin issues (acne, pigmentation, dark circles, wrinkles, enlarged pores). 4. The model identifies the skin tone and undertone. 5. The detailed analysis results are stored in the user's database record. 6. The system automatically proceeds to the "Skincare Recommendations" phase.
Alternative/Exception Flow	A1: Low Confidence: If the model's confidence is low, the system requests a new image or displays results with a caution regarding accuracy.
Postconditions	Analysis results are stored successfully; inputs for recommendation generation are determined.

TABLE 6 - AI SKIN ANALYSIS

5. Skincare Recommendations

Element	Description
Use Case Name	Skincare Recommendations
Goal	Provide a tailored daily skincare routine and suggest suitable products based on analysis results and gender differences.
Actor	System
Preconditions	AI analysis is complete, identifying skin type and conditions.
Basic Flow	1. The system retrieves the AI analysis results and the user's gender. 2. The system queries the product database for a suitable daily routine (AM/PM) based on skin type and conditions. 3. The system retrieves a list of recommended safe and effective products and brands. 4. Recommendations are customized to account for biological differences between men's and women's skin. 5. The recommended routine and product list are compiled for display.
Alternative/Exception Flow	A1: No Product Match: If specific products are unavailable regionally, the system recommends suitable active ingredients instead.
Postconditions	A personalized routine and product list are generated; the system is prepared for the "Results Display" use case.

TABLE 7 - SKINCARE RECOMMENDATIONS

6. Beauty Product Suggestions (for Women)

Element	Description
Use Case Name	Beauty Product Suggestions (for Women)
Goal	Provide tailored cosmetic recommendations, such as foundation or BB/CC cream, based on undertone and skin tone.
Actor	System
Preconditions	The user is female; skin tone and undertone have been identified by the AI analysis.
Basic Flow	1. The system confirms the user is female and retrieves the skin tone/undertone data. 2. The system queries the cosmetic product database. 3. The system matches the tone and undertone to the most suitable cosmetic shade (e.g., foundation, concealer). 4. The system presents a list of suggested cosmetic items and matching shades.
Alternative/Exception Flow	A1: User Preference: The user can opt out of receiving cosmetic suggestions in their profile settings.
Postconditions	Cosmetic recommendations are included in the final results for display.

TABLE 8 - BEAUTY PRODUCT SUGGESTIONS (FOR WOMEN)

7. Saving Results and Progress Tracking

Element	Description
Use Case Name	Saving Results and Progress Tracking
Goal	Allow users to monitor skin improvement over time by saving and comparing current and previous analysis results.
Actor	The User
Preconditions	A new skin analysis has been successfully performed.
Basic Flow	1. The system prompts the user to save the current analysis result. 2. The user confirms saving. 3. The system stores the analysis record (result, date, image) in the user's personal database. 4. The user can access a "Progress Tracking" view. 5. The system allows the user to select two records (e.g., current and oldest) to compare "before and after" results.
Alternative/Exception Flow	A1: User Declines Save: If the user declines, the record is not stored in the personal history.
Postconditions	The user's analysis history is updated; skin improvement is made trackable.

TABLE 9 - SAVING RESULTS AND PROGRESS TRACKING

8. Results Display

Element	Description
Use Case Name	Results Display
Goal	Clearly and systematically present the AI analysis results and all personalized recommendations to the user.
Actor	The User
Preconditions	AI analysis and recommendation generation are complete.
Basic Flow	1. The system receives all compiled data (analysis, routine, products, cosmetics). 2. The system displays the final results screen. 3. AI analysis findings (type, issues) are presented clearly, supporting both Arabic and English. 4. The system presents the recommended routine and product lists/shades. 5. The user is offered options to "Save Results" or "Start New Analysis."
Alternative/Exception Flow	A1: Missing Data: If recommendations could not be generated (e.g., database error), the system displays a partial result and an error message.
Postconditions	The user has received the required information; the analysis session is concluded.

TABLE 10 - RESULTS DISPLAY

9. Account or Data Deletion

Element	Description
Use Case Name	Account or Data Deletion
Goal	Allow the user to delete personal data, saved photos, or their entire account to comply with privacy and ethical standards.
Actor	The User
Preconditions	The user is logged into the system.
Basic Flow	1. The user navigates to the account settings. 2. The user selects the option to "Delete Data" or "Delete Account." 3. The system requests final confirmation for the irreversible action. 4. The user confirms the request. 5. The system securely removes all user-uploaded images and/or permanently deletes the user profile from the backend.
Alternative/Exception Flow	A1: User Cancels: The user stops the process, and the data remains saved.
Postconditions	User data is removed, ensuring confidentiality and adherence to security protocols.

TABLE 11 - ACCOUNT OR DATA DELETION

10. Rating and Review submission

Element	Description
Use Case Name	Rating and Review Submission
Goal	Allow users to rate their experience and submit written feedback on the skincare recommendations and analysis results.
Actor	The User
Preconditions	The user is logged in; an analysis session has been completed and its results have been displayed.
Basic Flow	1. The system displays a “Rate Your Experience” option on the results screen or within the saved analysis record. 2. The user selects a star rating (1–5). 3. The user optionally writes a text review describing their experience or satisfaction. 4. The user submits the rating and review. 5. The system stores the rating and review in the database linked to the user’s analysis record.
Alternative/Exception Flow	A1: No Text Review Provided: The user submits only a star rating; the system accepts and stores it. A2: User Cancels: The user exits the rating screen without submitting; no data is saved. A3: Connectivity Issue: If submission fails, the system displays an error message and prompts the user to retry.
Postconditions	The rating and optional review are successfully stored and become available for analytics and future system improvements.

TABLE 12 - RATING AND REVIEW SUBMISSION

5.8 System Class Diagram

The system is based on a set of organized categories designed to facilitate expansion and maintenance. This includes the basic categories of user, image analyzer, recommendation engine, and product category, with each representing an important part of the system's structure.

The user category includes personal data such as name, email, password, and gender, as well as account management functions. The image analyzer category is responsible for receiving uploaded images, pre-processing them, and then passing them to the artificial intelligence model for accurate analysis.

After the analysis is finished, the recommendation engine category uses the results to generate appropriate suggestions for skin care or cosmetics, drawing on product data stored in the product category, which includes detailed information about each product, such as its

name, category, and target skin type. The categories are integrated into a sequential workflow that begins with the user uploading the image, continues with its processing, and ends with the issuing of recommendations and retrieval of product data, making the design simple and flexible and helping in the future development of the system.

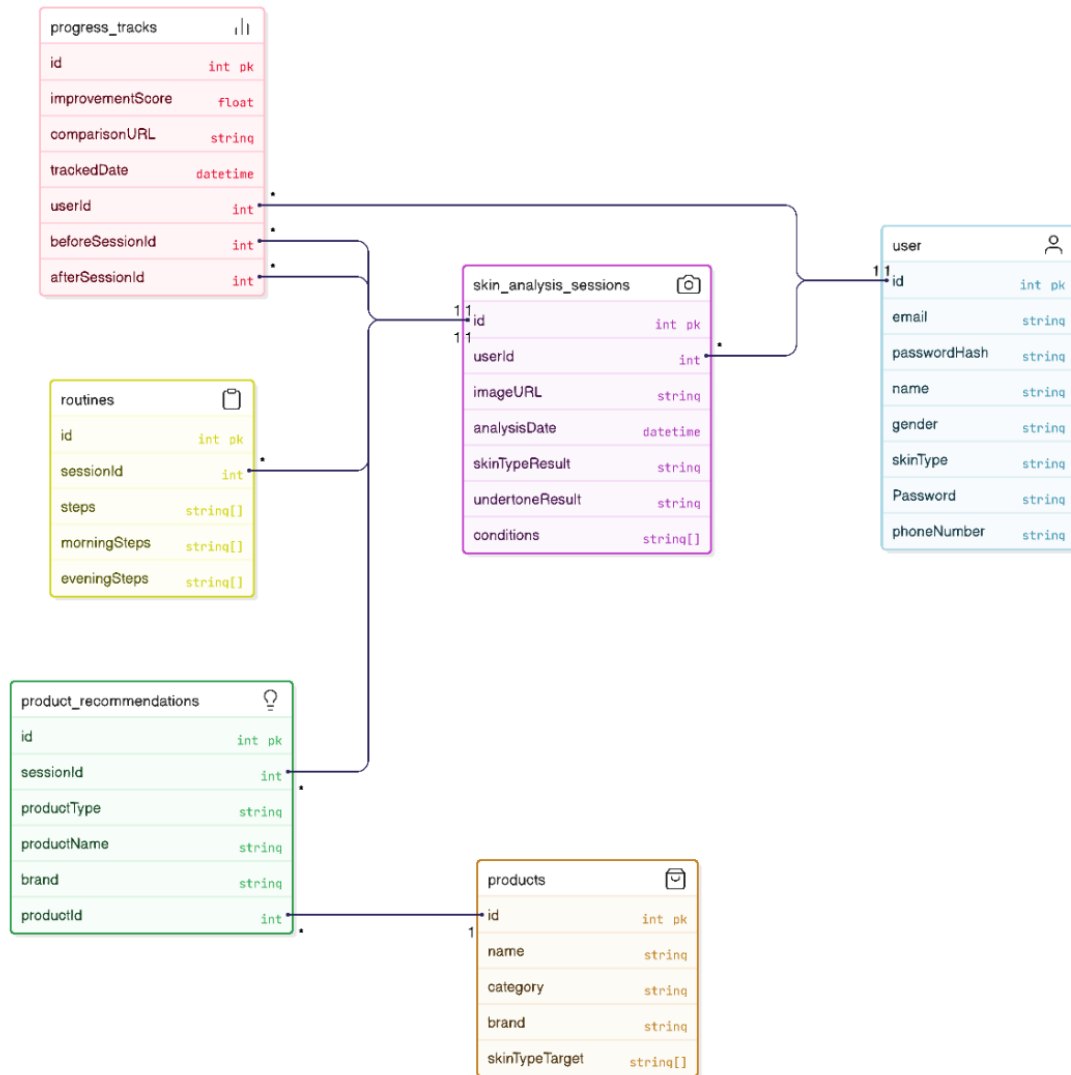


FIGURE 7 – SYSTEM CLASS DIAGRAM

5.9 System Interaction Diagram

In the Sequence diagram the process begins when the user initiates an analysis through the mobile app. The app processes the captured image locally, then sends a request to the backend controller for AI analysis. The backend forwards the image data to the AI model service, which analyzes facial features and returns the analysis results. These results are stored in the database.

Next, the backend requests personalized recommendations based on the analysis. The database returns the recommendations, and the backend sends the final combined results back to the mobile app, where they are displayed to the user.

Finally, the user submits a rating and review through the app. This feedback is sent to the backend, saved in the database, and a confirmation is returned to the app, indicating successful submission.

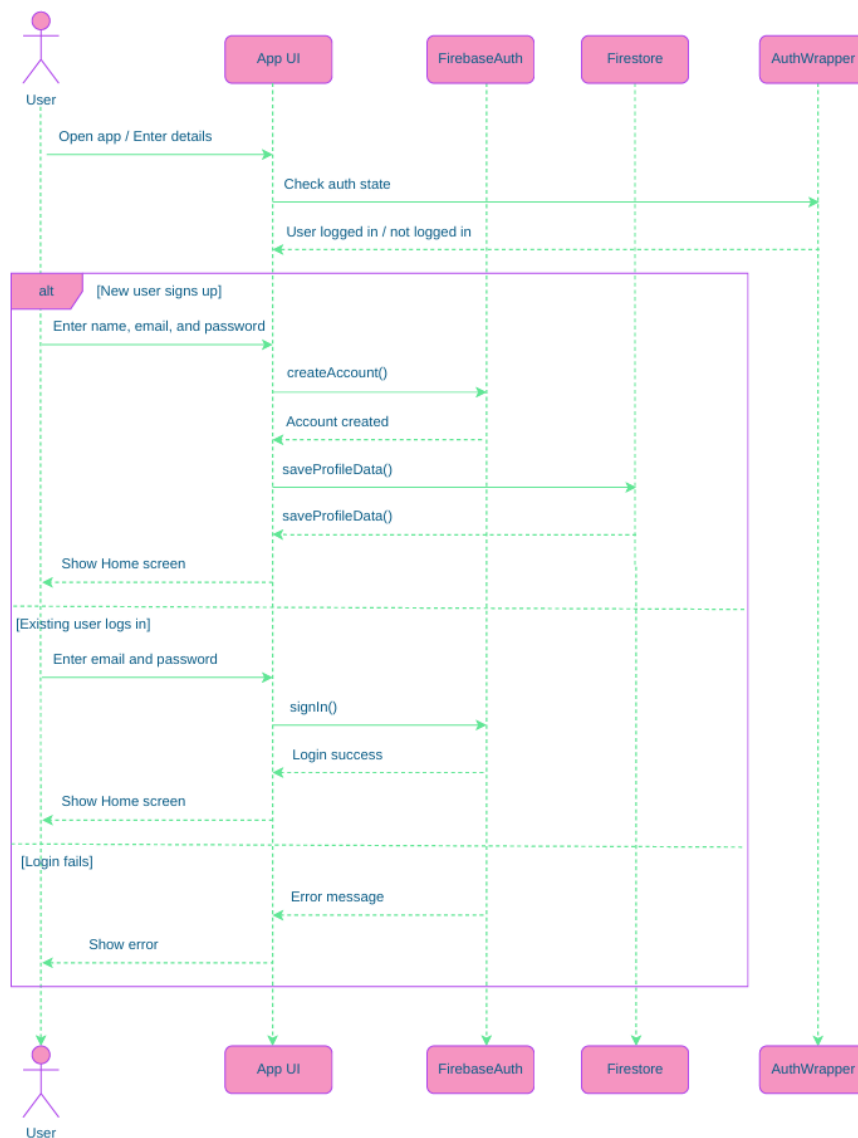


FIGURE 8 - REGISTRATION AND LOGIN

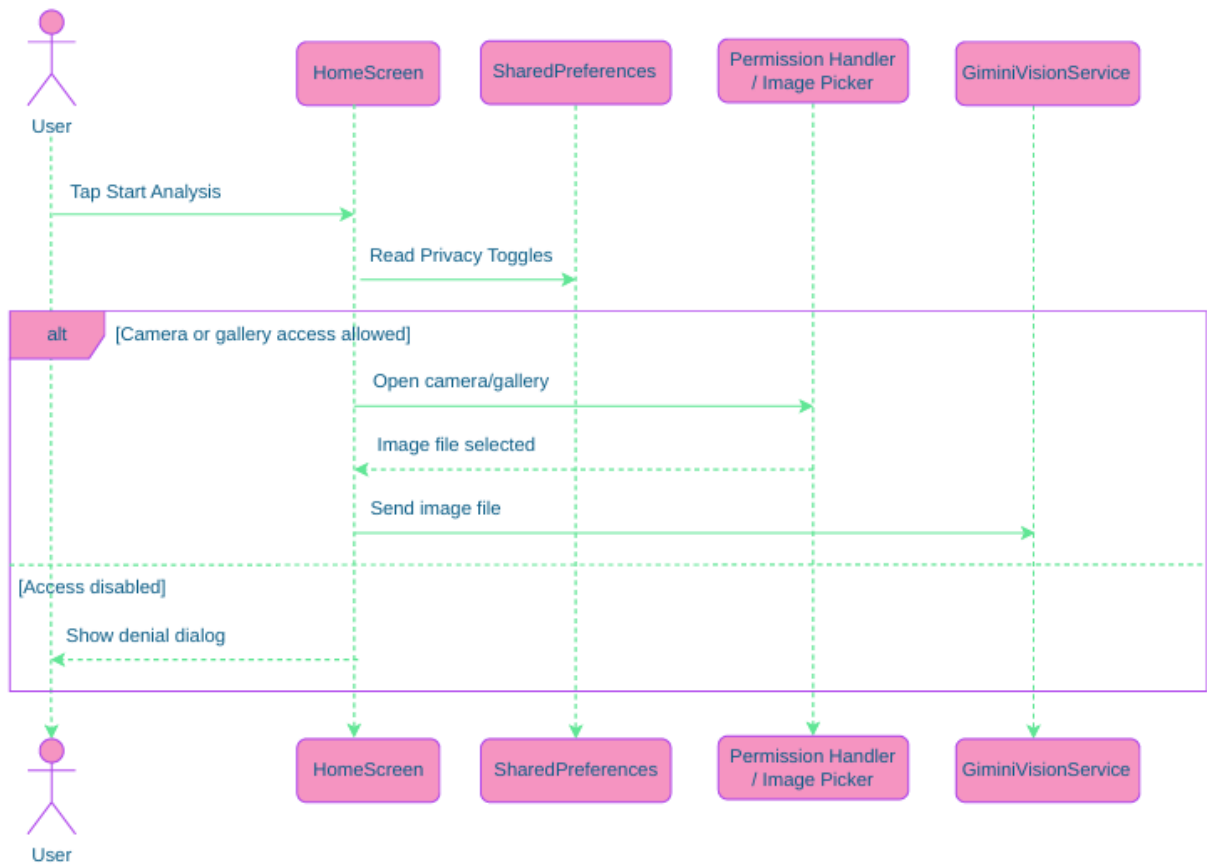


FIGURE 9 - ANALYSIS AND CAPTURE IMAGE

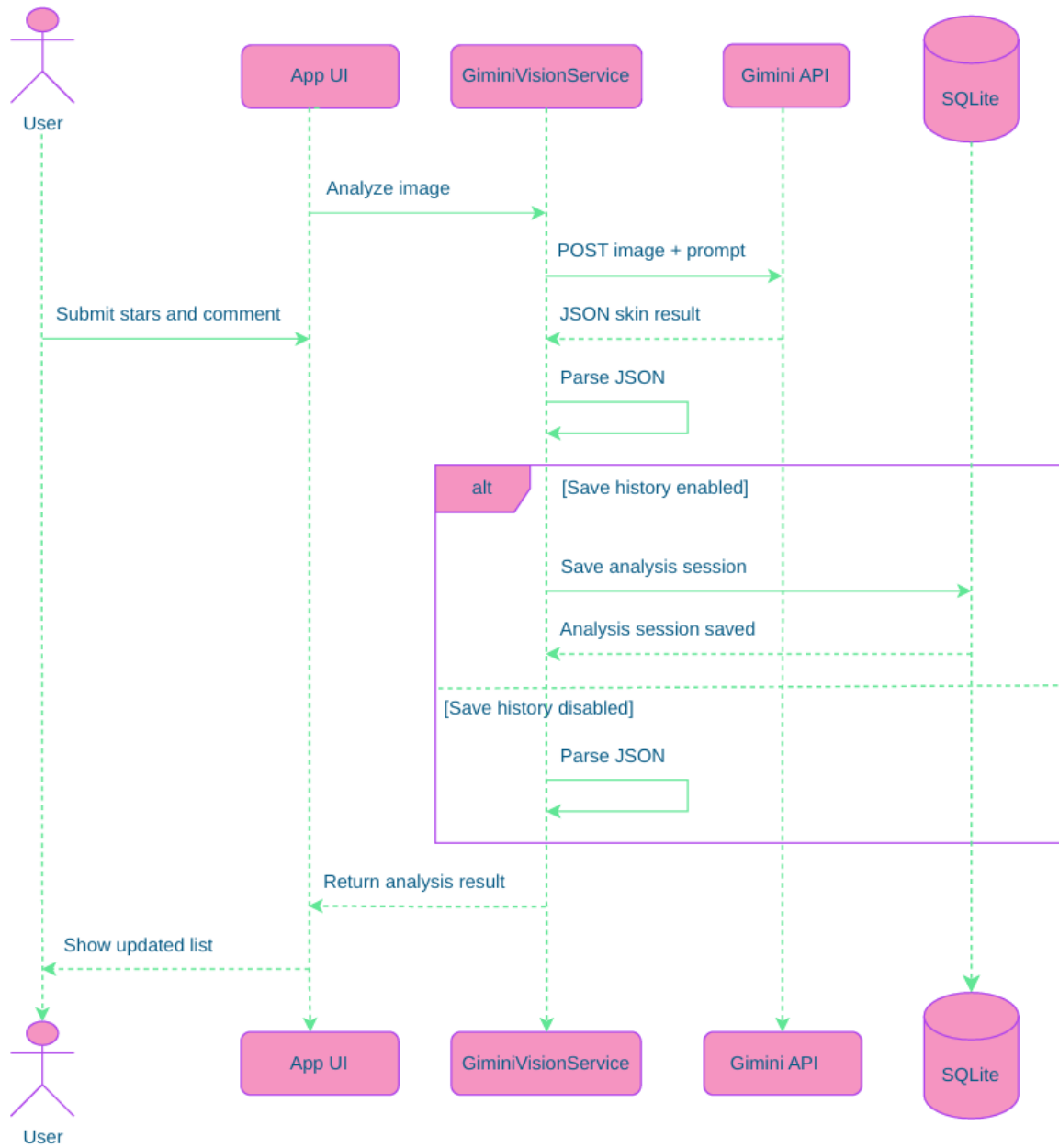


FIGURE 10 - AI SKIN ANALYSIS

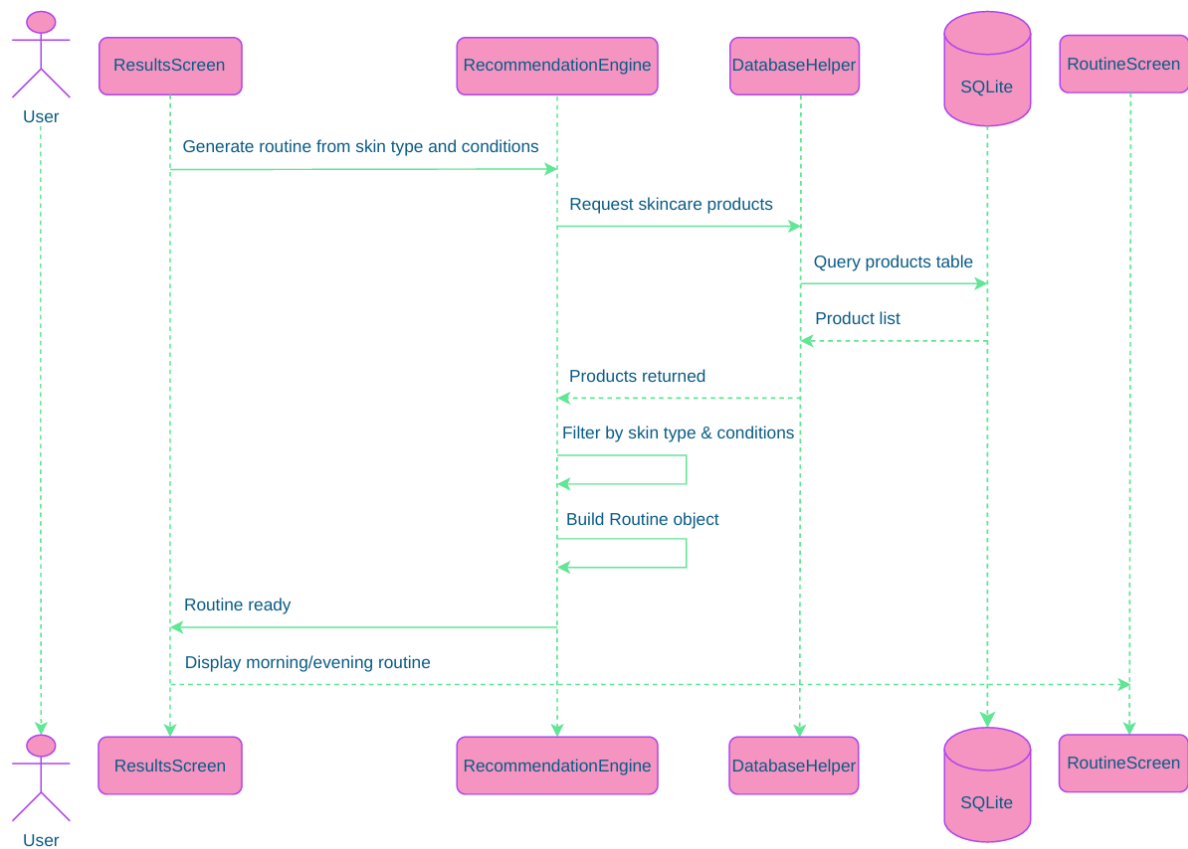


FIGURE 11 - ROUTINE AND SKINCARE RECOMMENDATIONS

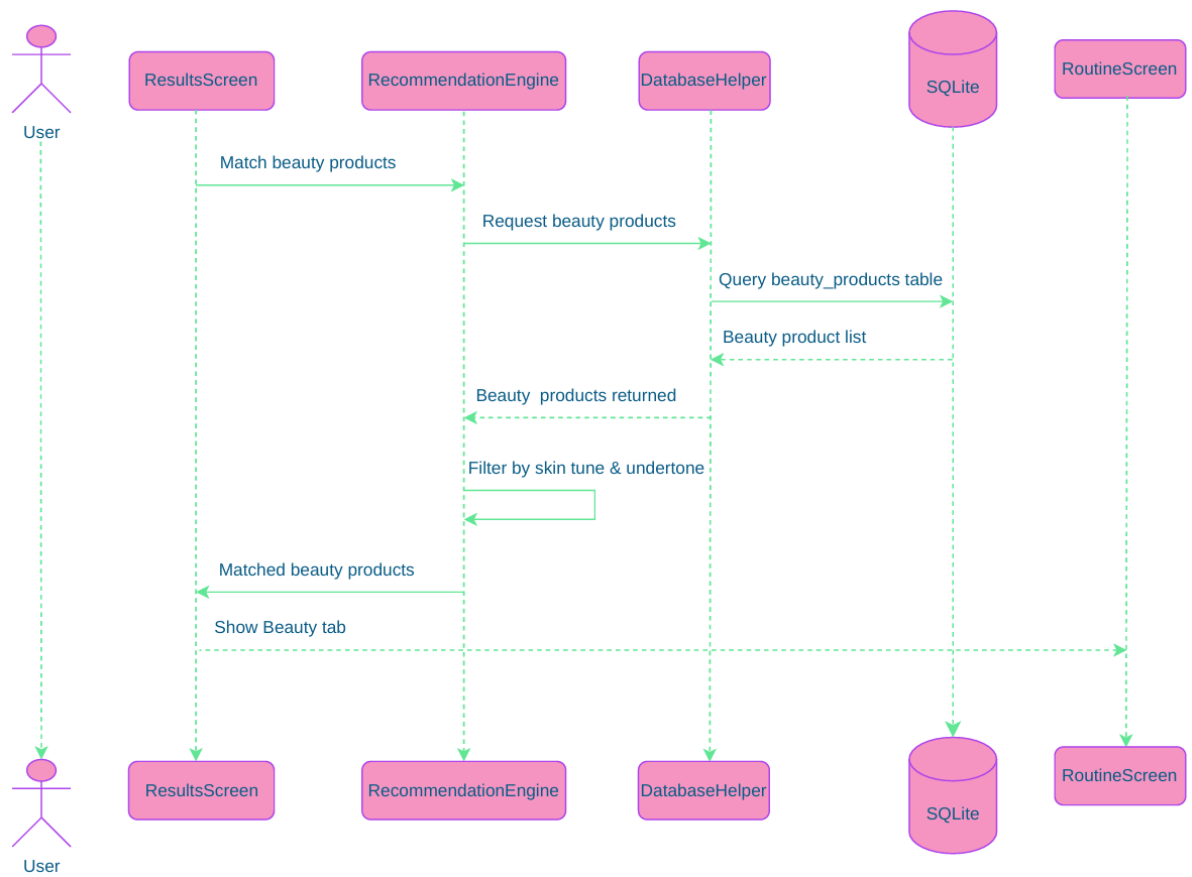


FIGURE 12 - BEAUTY PRODUCT MATCHING

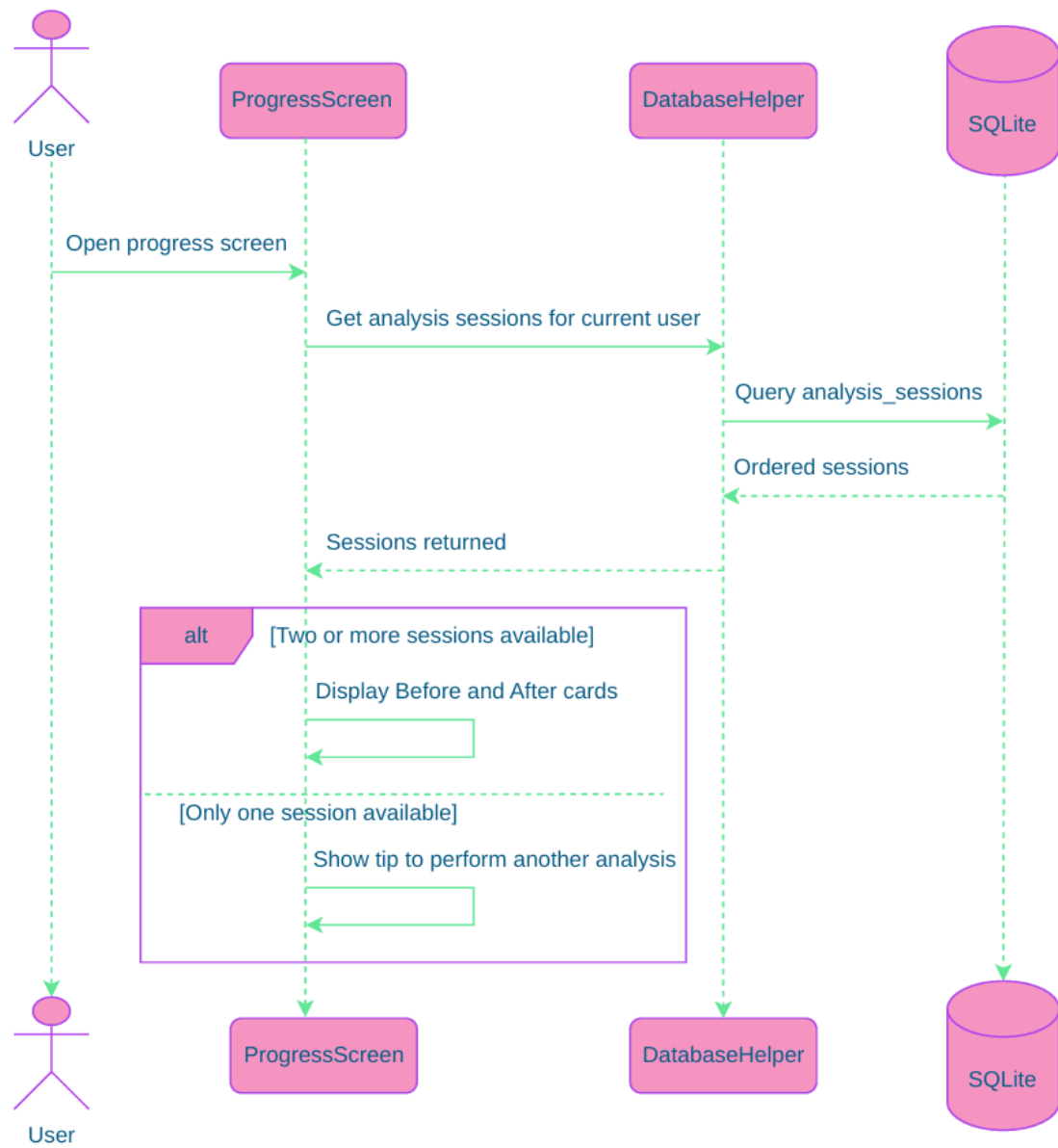


FIGURE 13 - PROGRESS TRACKING

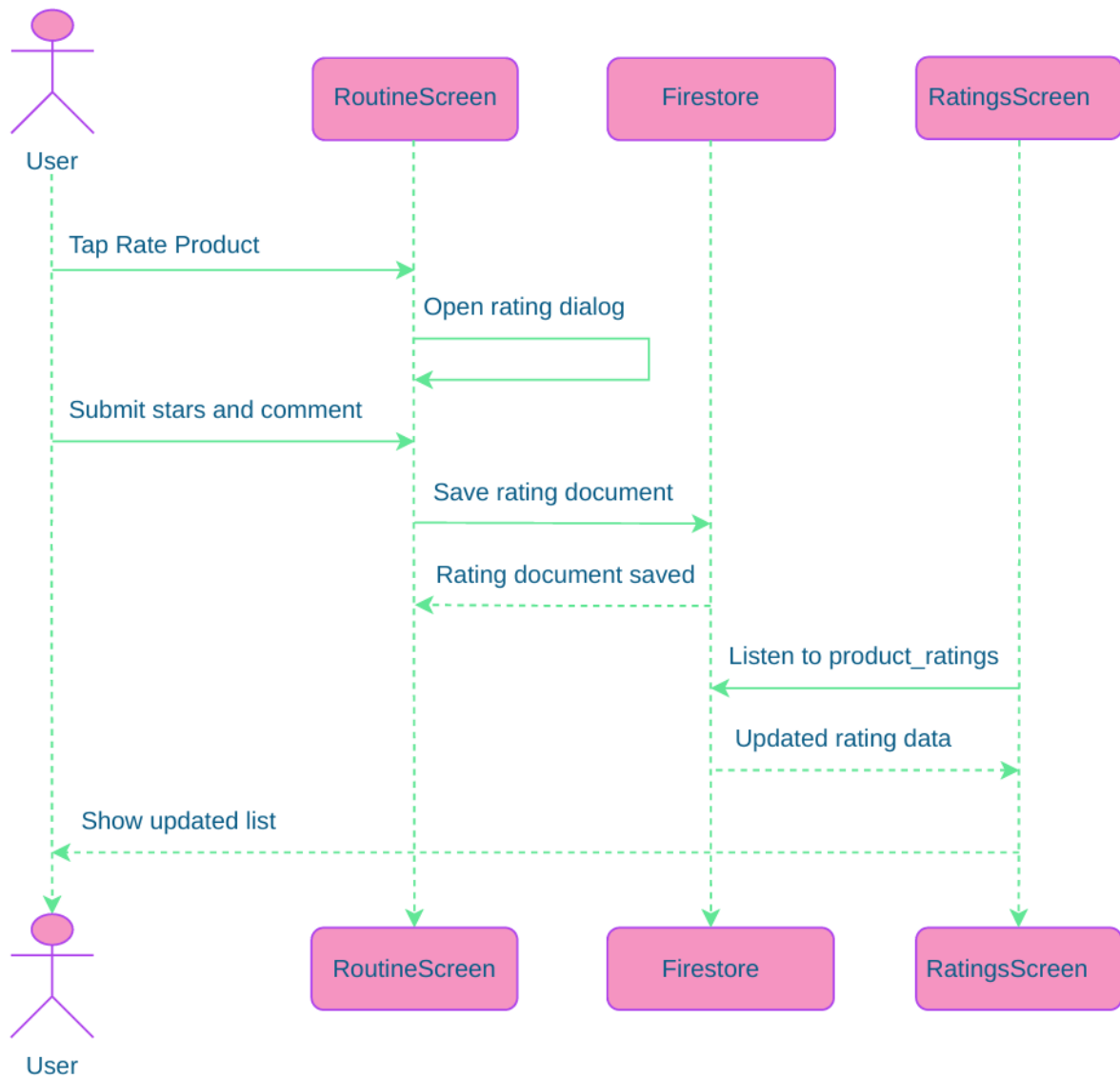


FIGURE 14 - PRODUCT RATING

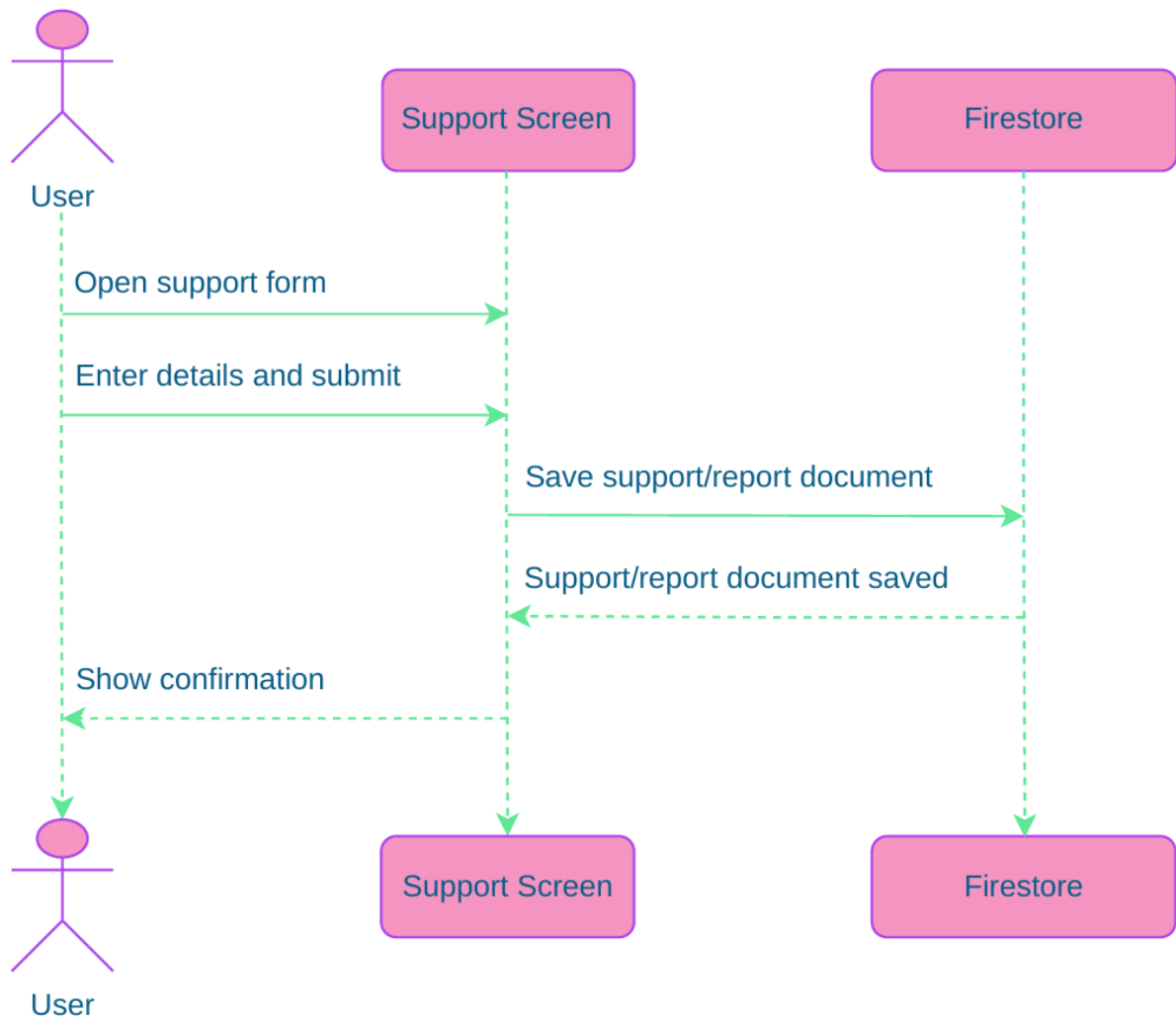


FIGURE 15 - REPORT PROBLEM AND CONTACT SUPPORT

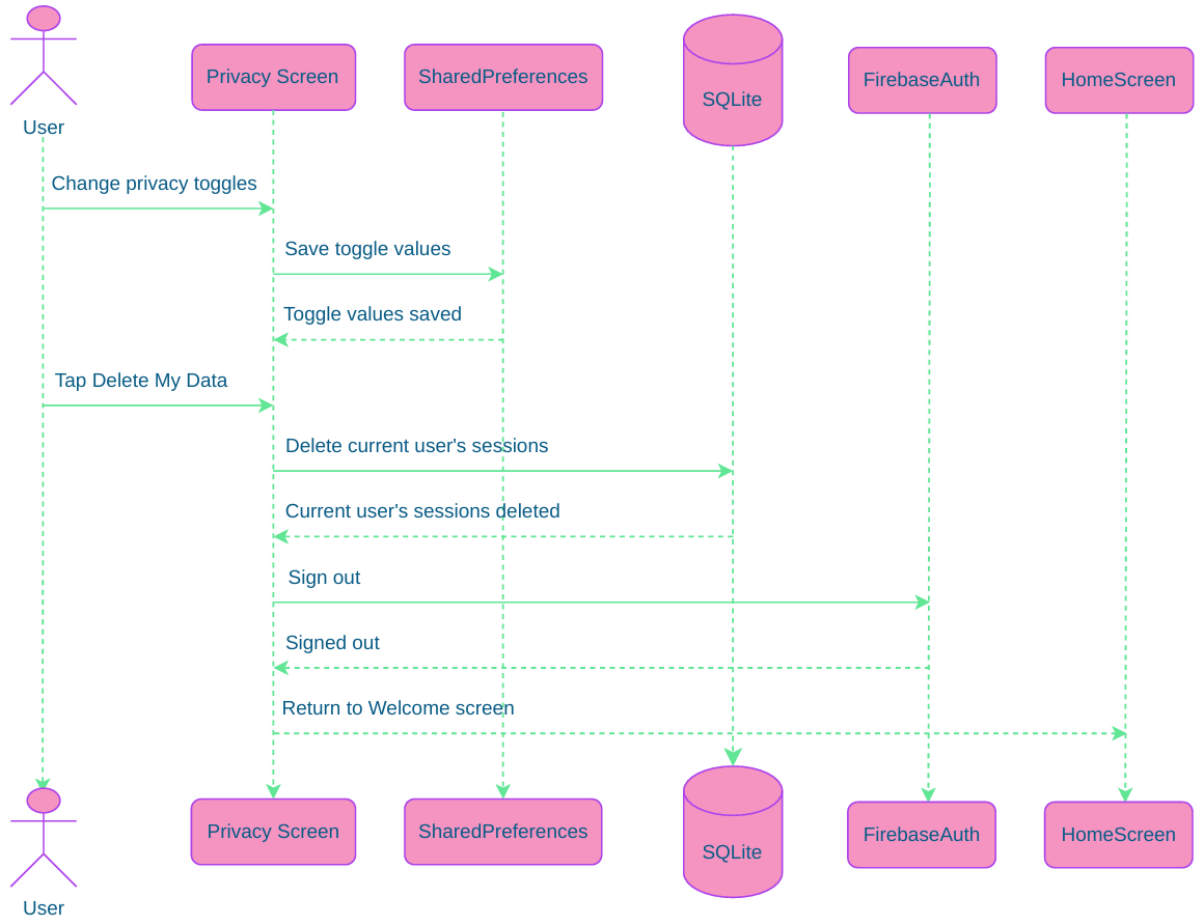


FIGURE 16 - PRIVACY SETTINGS AND DATA DELETION

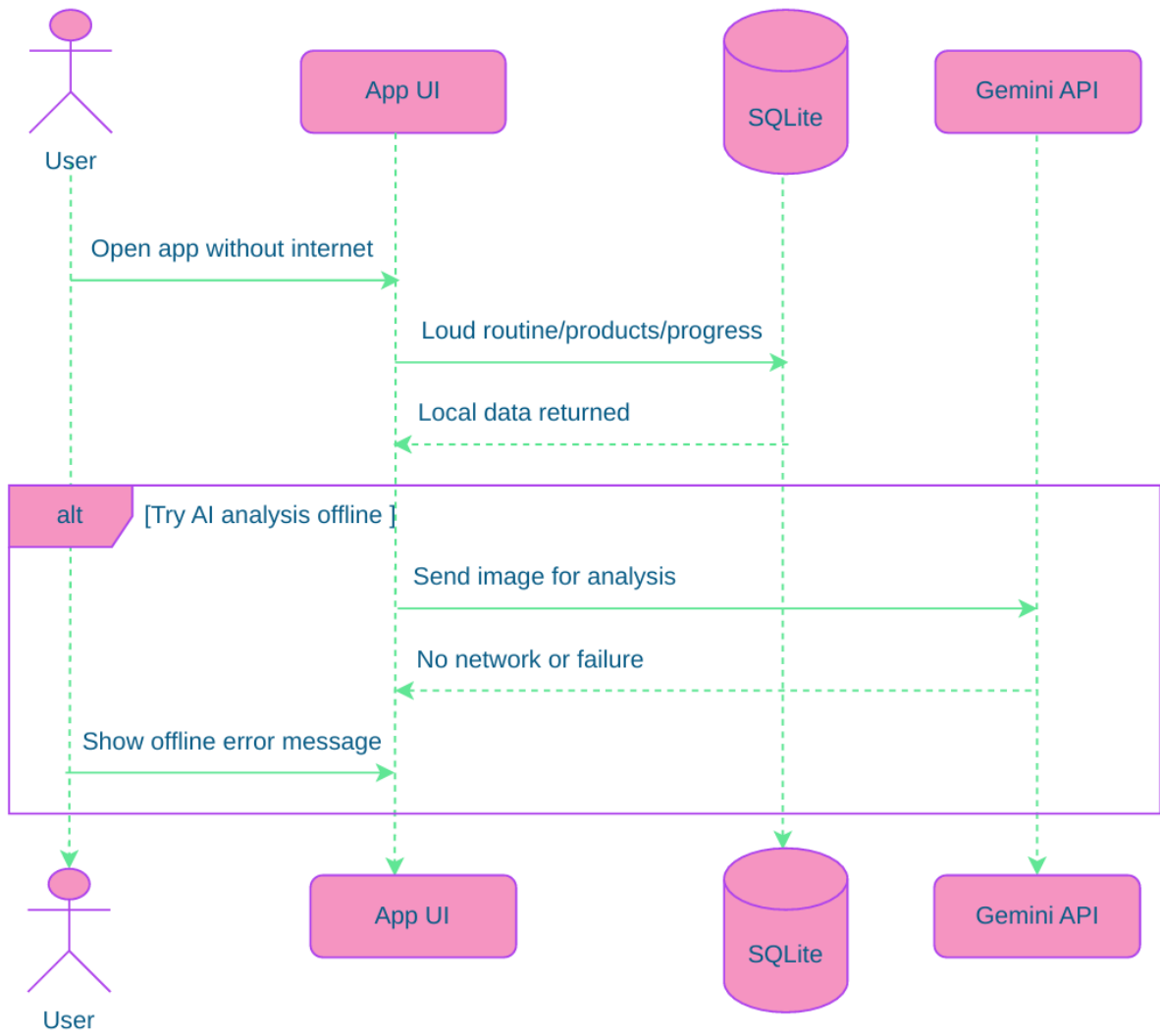


FIGURE 17 - OFFLINE ACCESS

5.10 System Activity Diagram

This is the activity diagram that models the typical flow of the user for the skin analysis feature. It consists of branches for image quality failure, user acceptance of recommendations, saving results, and feedback submission.

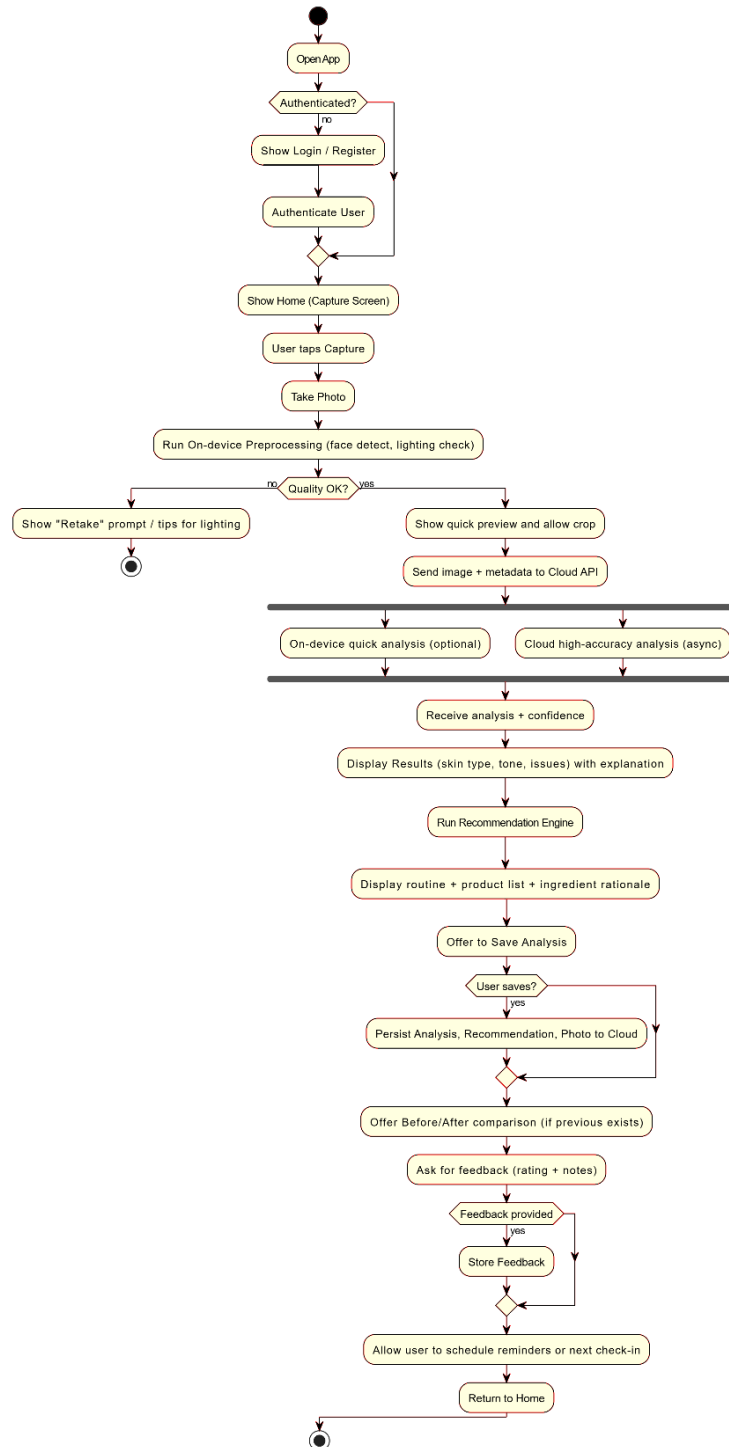


FIGURE 18 - SYSTEM ACTIVITY DIAGRAM

5.11 Front-End / User Interface Design

1. Design Philosophy and Usability Goals

Glow Up AI is focused on combining powerful AI with a smooth and reliable user experience.

- User-centered design: Simple, minimal, and straightforward interface with a streamlined process
- (capture or upload photo → analysis → display results and recommendations → evaluation and feedback).
- Transparency: Clear indications and explanations to build trust in AI decisions.
- Inclusivity: Supports both Arabic and English with clear and easily accessible images.

2. Key Interface Screens

2.1. Image Capture & Processing

- Visual guidelines for proper lighting and distance.
- Transparent progress bar showing automatic face detection and image enhancement before analysis.

2.2. AI Analysis Results

- Clear view of skin type and problems (acne, pigmentation, wrinkles, etc.) and color/base color.
- Fully bilingual labels and results.

2.3. Personalized Recommendations

- Personalized morning and evening skincare routines based on the conditions detected.
- Recommendations for products and brands tailored to skin type and gender.
- Makeup color matching for women.

2.4. Progress Tracking

- Before/after compare with dates, photos, and changes in metrics.

3. Usability and Security

- Security: Clear privacy communication and easy option to delete data.
- Reliability: Friendly prompts for unclear photos or communication issues.
- Modularity: Separate analysis and recommendation components to facilitate future updates.

5.12 Database / Data Model Design

This subsection documents the details of the logical model for **Glow Up AI** data.

It uses the Firestore storage system (or real-time database) for organized records and

Firebase Storage for images and larger files. The following ER model focuses on the core entities and relationships: user, image, analysis, recommendation, product, and comments.

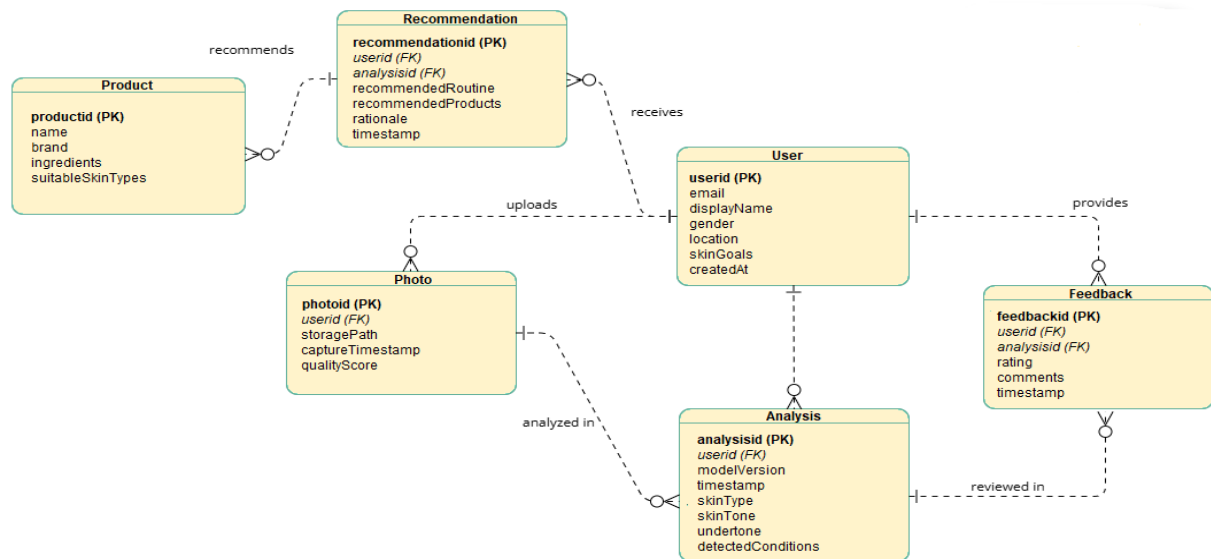


FIGURE 19 - DATA MODEL DESIGN

5.13 Algorithm / Process Design

The **Glow Up AI** system is based on a structured algorithm that analyzes the user's skin and provides personalized recommendations tailored to their needs. This process is designed to be efficient, transparent, and easy to maintain.

The basic steps of this algorithm are as follows:

1. The user takes a photo of their face or uploads one via the app interface.
2. The system performs a series of preliminary processing operations, such as adjusting the lighting, cropping the image, and standardizing its characteristics to ensure the quality of the analysis.
3. The processed image is passed to a convolutional neural network (CNN) model to extract and classify features.

4. The model determines skin type, color, and any visible imperfections.
5. The analysis results are sent to the recommendation engine.
6. The recommendation engine matches the analysis results with skincare routines and products available in the database.
7. The final recommendations are stored with the analysis details and presented to the user in an accessible format.

The design aims to provide accurate and consistent analysis while maintaining a simplified workflow suitable for mobile devices. The system architecture has also been designed to be flexible and scalable, facilitating integration with other systems and platforms in the future.

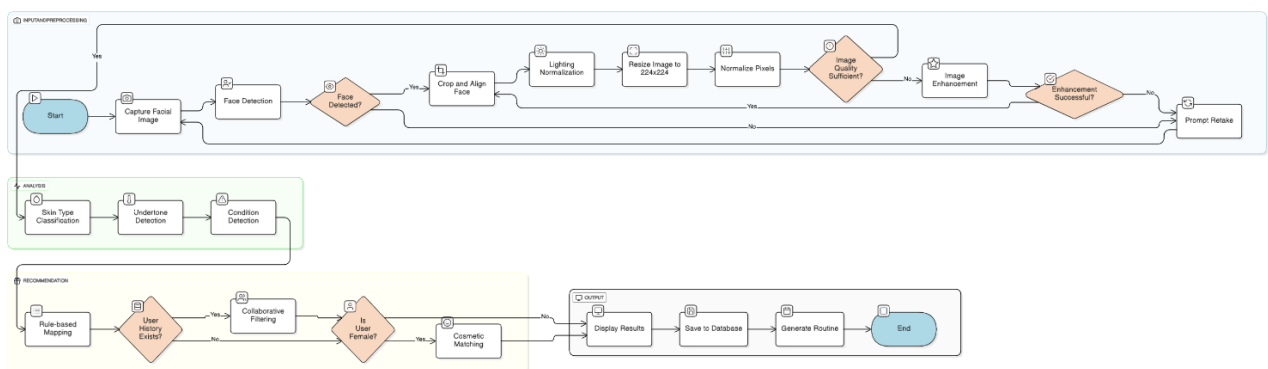


FIGURE 20 - ALGORITHM / PROCESS DESIGN

5.14 Summary

This section outlines the overall design of the **Glow Up AI** system and details how it works from the moment the user captures or uploads a face image to the moment the personalized results are displayed. It highlights the design methodology, the path that data follows through the system, and the key components of the architecture, including the mobile application interface, image preprocessing, AI-based analysis, and the recommendation module.

The chapter also includes basic diagrams such as a use case diagram, database diagram, and class structure, which visually explain how the system works and how data is organized. Furthermore, it describes the layout and features of the user interface, as well as the sequence of actions that the system performs to process images and provide personalized skincare advice.

Overall, this section lays the necessary groundwork for the next chapter, which focuses on the step-by-step implementation of the system.

Chapter 6

IMPLEMENTATION / EXPERIMENTATION

6.1 Implementation Plan

The Glow Up AI system was developed using the Agile methodology. The project was divided into six sprints over a period of 12 weeks to ensure gradual development, continuous testing, and regular improvements throughout the implementation process.

The main technologies used during development were:

- Flutter for mobile application development.
- Dart as the programming language.
- Android Studio as the development environment.
- Firebase Authentication for user management.
- Cloud Firestore for cloud data storage.
- SQLite for local data storage.
- Google Gemini API for AI-powered skin analysis.

Sprint	Duration	Tasks
Sprint 1	Weeks 1–2	Project setup, Firebase configuration, dependency installation, and project structure preparation.
Sprint 2	Weeks 3–4	Development of user authentication and account management features.
Sprint 3	Weeks 5–6	SQLite database design and skincare and beauty product catalogue development.
Sprint 4	Weeks 7–8	Integration of Gemini AI and implementation of skin analysis functionality.
Sprint 5	Weeks 9–10	Development of the recommendation engine, skincare routines, beauty matching, and progress tracking.
Sprint 6	Weeks 11–12	System testing, bug fixing, performance validation, and final deployment preparation.

TABLE 13 - SPRINT SCHEDULE

The implementation plan ensured that each component was completed, tested, and refined before moving to the next stage.

6.2 Module / Process Development

The **Glow Up AI** system was developed through several interconnected modules. Each module was implemented individually and then integrated into the complete system.

6.2.1 Authentication Module

The authentication module was implemented using Firebase Authentication. It enables users to create accounts, log in, log out, and recover forgotten passwords. User information is securely stored and linked to each account, ensuring personalized access to application features.

6.2.2 Local Database Module

SQLite was used as the local database solution to support offline access and improve application performance. The local database stores:

- Skincare product catalogues.
- Beauty product catalogues.
- User analysis history.
- Progress tracking information.

This approach allows users to access important information even when internet connectivity is unavailable.

6.2.3 Experimental Custom AI Model Development

Before integrating Google Gemini API, an experimental custom AI model was developed and evaluated as part of the project implementation process. The purpose of this model was to classify several skin-related attributes from generated image samples.

6.2.3.1 Synthetic Dataset Generation

A synthetic dataset was generated programmatically using Python. The dataset generation process created image samples representing different combinations of skin-related characteristics.

The generated dataset included:

- Skin types (Oily, Dry, Combination, and Normal).
- Skin tones (Light, Medium, and Dark).
- Undertones (Warm, Cool, and Neutral).
- Gender categories (Male and Female).
- Acne status (Present or Absent).

A total of 2,000 images were generated and stored with their corresponding metadata labels in JSON format. Each image was created with a resolution of 224×224 pixels.

6.2.3.2 Model Architecture

The model was implemented using TensorFlow and Keras based on a Convolutional Neural Network (CNN) architecture. The network receives input images with dimensions of $224 \times 224 \times 3$ and processes them through three convolutional layers containing 32, 64, and 128 filters respectively. Each convolutional layer is followed by a max-pooling layer to reduce the spatial dimensions of the extracted features. The resulting feature maps are then flattened and passed to a fully connected dense layer consisting of 512 neurons, followed by a dropout layer to support the training process.

The architecture was designed as a multi-output classification model with five output branches: type_out, tone_out, undertone_out, problem_out, and gender_out. These outputs were used to predict the corresponding attributes from the generated image samples. The complete model contained a total of 44,404,304 trainable parameters.

Model: "functional_1"

...

Layer (type)	Output Shape	Param #	Connected to
face_input (InputLayer)	(None, 224, 224, 3)	0	-
conv2d_3 (Conv2D)	(None, 222, 222, 32)	896	face_input[0][0]
max_pooling2d_3 (MaxPooling2D)	(None, 111, 111, 32)	0	conv2d_3[0][0]
conv2d_4 (Conv2D)	(None, 109, 109, 64)	18,496	max_pooling2d_3[...]
max_pooling2d_4 (MaxPooling2D)	(None, 54, 54, 64)	0	conv2d_4[0][0]
conv2d_5 (Conv2D)	(None, 52, 52, 128)	73,856	max_pooling2d_4[...]
max_pooling2d_5 (MaxPooling2D)	(None, 26, 26, 128)	0	conv2d_5[0][0]
flatten_1 (Flatten)	(None, 86528)	0	max_pooling2d_5[...]
dense_1 (Dense)	(None, 512)	44,302,848	flatten_1[0][0]
dropout_1 (Dropout)	(None, 512)	0	dense_1[0][0]
type_out (Dense)	(None, 4)	2,052	dropout_1[0][0]
tone_out (Dense)	(None, 3)	1,539	dropout_1[0][0]
undertone_out (Dense)	(None, 3)	1,539	dropout_1[0][0]
problem_out (Dense)	(None, 4)	2,052	dropout_1[0][0]
gender_out (Dense)	(None, 2)	1,026	dropout_1[0][0]

Total params: 44,404,304 (169.39 MB)
Trainable params: 44,404,304 (169.39 MB)
Non-trainable params: 0 (0.00 B)

FIGURE 21: CNN MODEL ARCHITECTURE SUMMARY

6.2.3.3 MODEL EVALUATION

After training, the model was evaluated using the generated dataset. The evaluation process produced performance metrics for all output branches, including skin type, skin tone, undertone, skin condition, and gender classification. The obtained results showed varying levels of accuracy across the different prediction tasks.

The detailed evaluation metrics, accuracy values, and loss values are presented in the following figures.

SUB-METRIC ACCURACY DETAILS		
	Metric	Final Accuracy
0	gender_out_accuracy	0.5019
1	problem_out_accuracy	1.0000
2	tone_out_accuracy	1.0000
3	type_out_accuracy	0.2519
4	undertone_out_accuracy	0.3219
5	val_gender_out_accuracy	0.5025
6	val_problem_out_accuracy	1.0000
7	val_tone_out_accuracy	1.0000
8	val_type_out_accuracy	0.2475
9	val_undertone_out_accuracy	0.3125

GLOBAL MODEL ACCURACY PERFORMANCE
Global Model Accuracy: 0.6138 (61.38%)

FIGURE 22:CNN MODEL ACCURACY PERFORMANCE METRICS

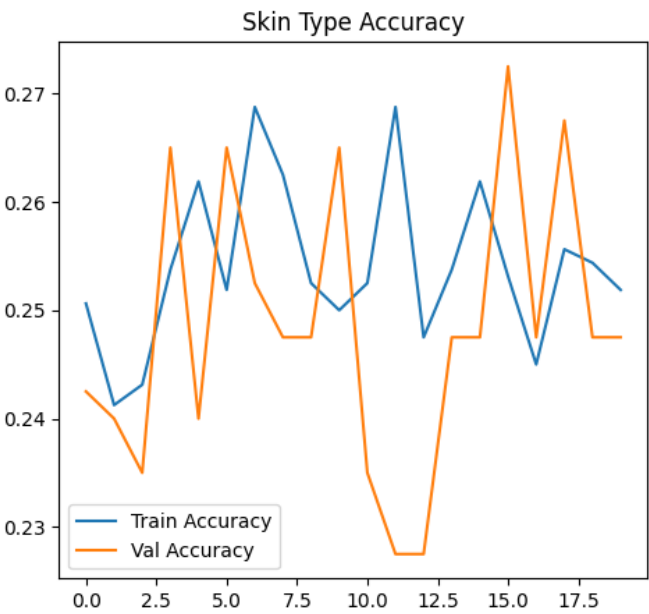


FIGURE 23:MODEL ACCURACY EVALUATION RESULTS

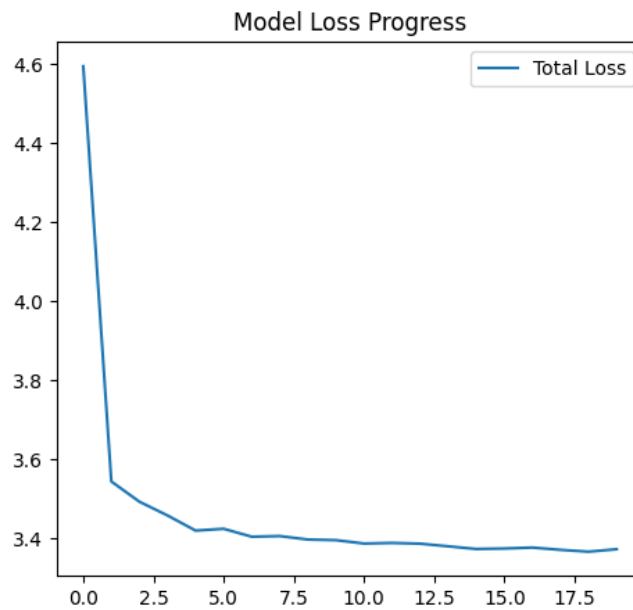


FIGURE 24:MODEL LOSS PROGRESS DURING TRAINING

6.2.3.4 SAMPLE PREDICTION RESULT

To demonstrate the output of the trained model, a sample image was analyzed during the testing phase. The model generated predictions for the available classification categories and produced the results shown in the following figure.

```
Prediction Summary:
- Skin Type: Combination (26.2%)
- Skin Tone: Fair (57.8%)
- Skin Condition Probability: 49.1% -> Clear
=====
No matching recommendations found.
```

FIGURE 25:SAMPLE PREDICTION RESULT GENERATED BY THE CNN MODEL

6.2.3.5 Transition to Gemini API

Following the evaluation of the experimental CNN model, the obtained results indicated that the model did not provide the level of performance required for the intended application. Therefore, an alternative AI solution was considered for the skin analysis component.

After investigating available options, Google Gemini API was selected and integrated into the system. The final implementation of **Glow Up AI** utilizes Gemini API as the primary AI service for skin analysis and recommendation generation.

6.2.4 AI Skin Analysis Module

The AI analysis module utilizes Google Gemini API to analyze user facial images. The system identifies key skin characteristics, including:

- Skin type.
- Skin tone.
- Undertone.
- Skin conditions such as acne, dryness, redness, and pigmentation.

The analysis results are presented to the user and can be stored locally for future reference.

6.2.5 Recommendation Engine Module

The recommendation engine generates personalized skincare and beauty recommendations based on the analysis results.

The recommendation process includes:

1. Identifying the user's skin type.
2. Detecting skin conditions.
3. Matching skincare products to the user's needs.
4. Generating a personalized morning routine.
5. Generating a personalized evening routine.
6. Recommending beauty products based on skin tone and undertone.

This module provides customized suggestions designed to improve the user's skincare experience.

6.2.6 Progress Tracking Module

The progress tracking module enables users to monitor changes in their skin condition over time. Previous analysis sessions are stored and displayed in a comparison format, allowing users to observe improvements and track their skincare journey.

6.2.7 Additional Support Modules

Several supporting modules were implemented to enhance the overall user experience, including:

- Product ratings and reviews.
- Contact support functionality.
- Problem reporting functionality.
- FAQ section.
- Privacy and permissions settings.

These modules contribute to a more complete and user-friendly application.

6.3 Integration of Components

After the development of individual modules, integration was performed to ensure that all system components worked together correctly.

The authentication system was connected to the user interface to control access to application features. The AI analysis module was integrated with the recommendation engine so that analysis results could be used directly to generate skincare routines and beauty product suggestions.

The local database was connected to the progress tracking module to support offline access to previous analysis sessions. Firestore was integrated with ratings, support messages, and problem reports to ensure secure cloud storage and synchronization.

Integration testing included the following end-to-end scenarios:

- User registration.
- User login and logout.
- Skin analysis.
- Routine generation.
- Progress tracking.
- Product rating submission.
- Support message submission.
- Problem report submission.
- Privacy settings management.
- User data deletion.

All integration scenarios were completed successfully, confirming the correct interaction between system components.

6.4 User Interface / Backend Development

6.4.1 User Interface Development

The user interface was developed using Flutter and Material Design principles to provide a simple, modern, and user-friendly experience.

Screen	Description
Welcome Screen:	Entry point of the application.
Terms and Privacy Policy Screen:	Displays terms and privacy information
Login Screen:	Allows users to access their accounts.
Sign Up Screen:	Allows new users to create accounts.
Forgot Password Screen:	Supports password recovery.
Home Screen;	Main navigation screen.
Skin Analysis Screen:	Allows users to upload or capture images.
Results Screen:	Displays AI analysis results.
Routine Screen:	Displays personalized skincare routines.
Beauty Products Screen:	Displays makeup recommendations.
Progress Tracking Screen:	Displays historical analysis comparisons
Ratings and Reviews Screen:	Displays user reviews and ratings.
Report Problem Screen:	Allows users to submit problem reports.
Contact Support Screen:	Allows users to contact support.
Settings Screen:	Contains privacy and application settings.

TABLE 14-MAIN APPLICATION SCREENS

The interfaces were designed to support smooth navigation and clear access to all application features.

6.4.2 Backend Development

The backend infrastructure combines both cloud and local storage solutions.

Cloud Firestore was used for:

- User profiles.
- Product ratings and reviews.
- Support messages.
- Problem reports.

SQLite was used for:

- Product catalogues.
- Analysis history.

- Progress tracking data.

Google Gemini API was integrated as the external AI service responsible for skin analysis and recommendation generation.

6.5 Deployment or Execution

The final version of **Glow Up AI** was installed and executed on a physical Android device for testing and demonstration purposes.

The deployment process included:

- Firebase configuration.
- SQLite database initialization.
- Gemini API integration.
- Application installation on a physical device.
- Functional testing of all major features.

The following activities were successfully performed during execution:

- User registration and login.
- Skin analysis using multiple images.
- Recommendation generation.
- Progress tracking.
- Product rating submission.
- Support message submission.
- Problem report submission.
- Privacy settings testing.

The application operated successfully without major errors and demonstrated stable performance during testing.

6.6 Summary

This chapter presented the implementation of the **Glow Up AI** system, including the development plan, system modules, integration process, user interface development, backend implementation, and deployment activities.

The system was developed using an Agile methodology over a period of 12 weeks and consisted of multiple integrated modules, including authentication, local data storage, AI skin analysis, recommendation generation, progress tracking, and support services.

Following successful implementation and integration, the system was prepared for testing and evaluation. The next chapter presents the results of functional testing, performance evaluation, usability observations, comparative analysis, challenges, limitations, and overall system assessment.

Chapter 7

RESULTS AND EVALUATION

7.1 Results Presentation

This chapter presents the outcomes of the **GlowUp AI** system after full implementation and testing. It includes the presentation of system results, functional testing outcomes, performance evaluation, a preliminary usability evaluation based on user experiences, comparative analysis, discussion of findings, challenges encountered, limitations, and a summary.

7.1.1 Application Interface Overview

The **Glow Up AI** application was successfully implemented with a complete set of interfaces that support the user journey from authentication to skincare and beauty recommendations. The interfaces were designed to provide a clear and user-friendly experience while maintaining smooth navigation between features.

The implemented interfaces include:

- Welcome Screen
- Login Screen
- Sign Up Screen
- Forgot Password Screen
- Home Screen
- Skin Analysis Screen
- Results Screen
- Routine Screen
- Beauty Products Screen
- Progress Tracking Screen
- Ratings and Reviews Screen
- Report Problem Screen
- Contact Support Screen
- Settings Screen

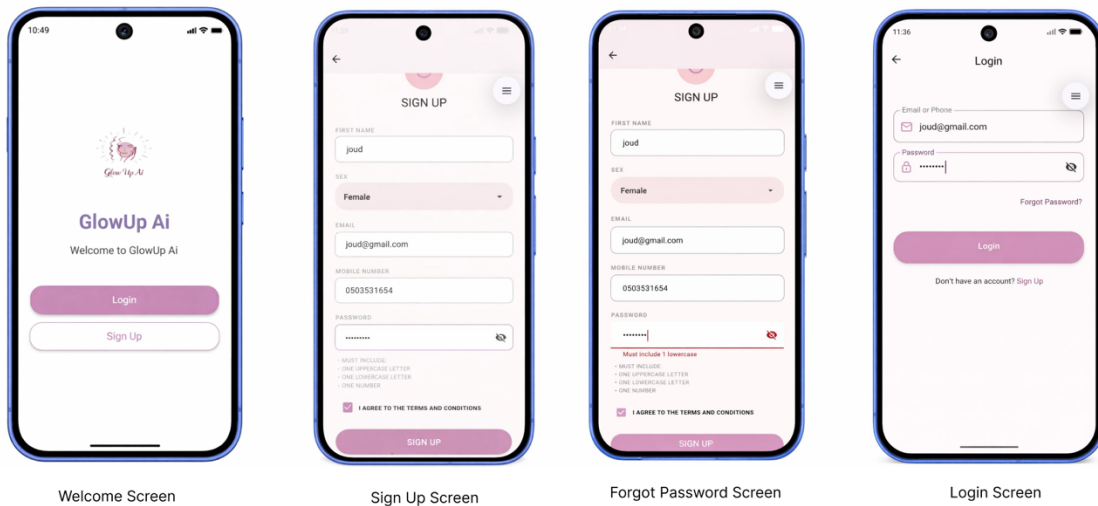


FIGURE 26:USER AUTHENTICATION SCREENS OF THE GLOWUP AI APPLICATION.

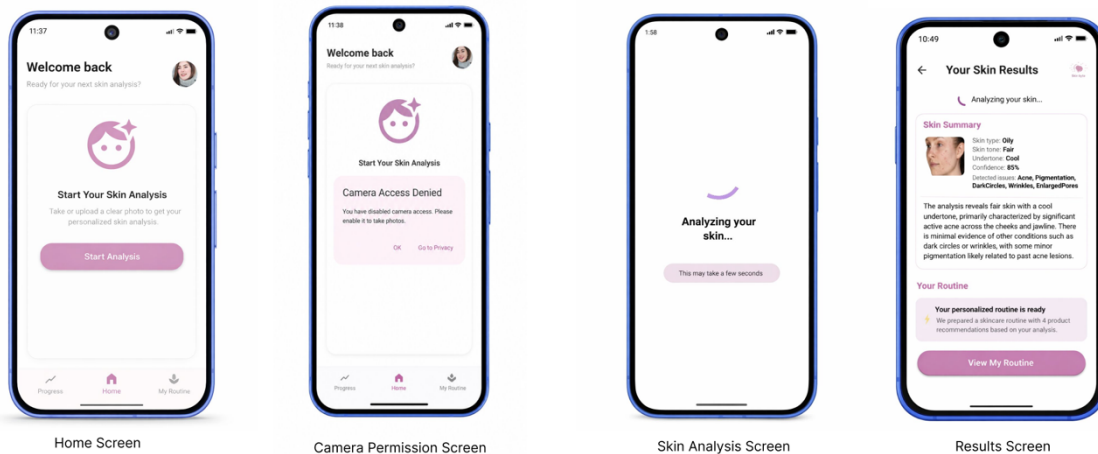
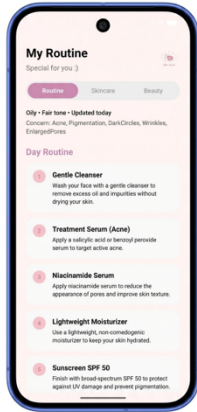
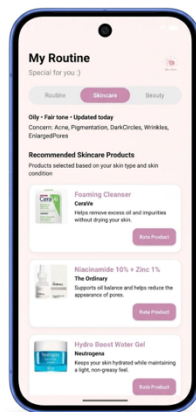


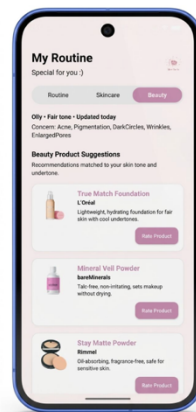
FIGURE 27:SKIN ANALYSIS AND RESULTS SCREENS OF THE GLOWUP AI APPLICATION.



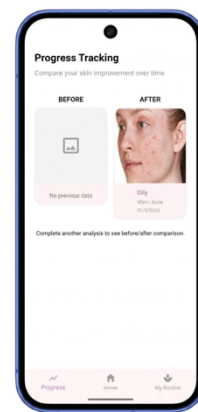
Routine Screen



Skincare Products Screen

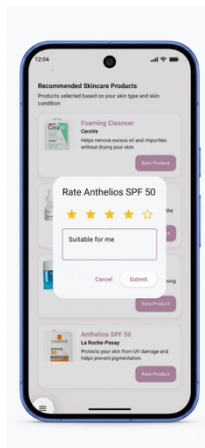


Beauty Products Screen

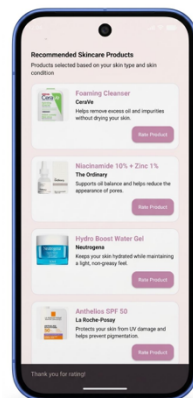


Progress Tracking Screen

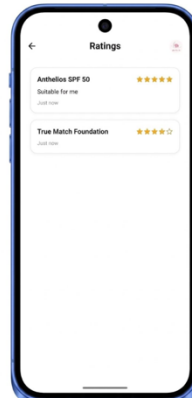
FIGURE 28:PERSONALIZED ROUTINE, PRODUCT RECOMMENDATION, AND PROGRESS TRACKING SCREENS OF THE GLOWUP AI APPLICATION.



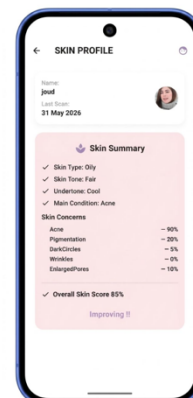
Ratings and Reviews Screen



Ratings and Reviews Screen



Ratings Screen



Skin Profile Screen

FIGURE 29:USER FEEDBACK, PRODUCT RATINGS, AND SKIN PROFILE SCREENS OF THE GLOWUP AI APPLICATION.

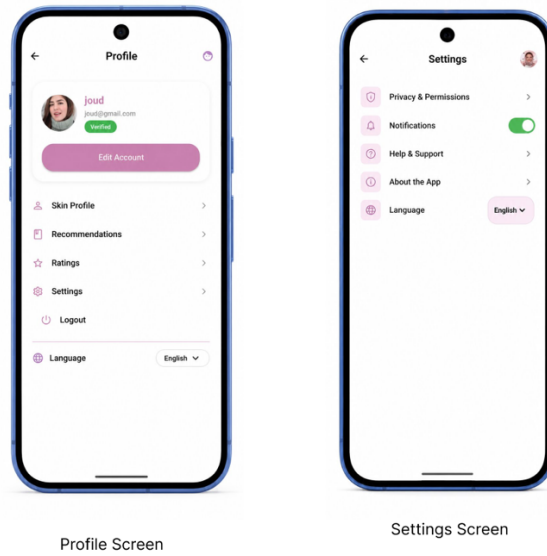


FIGURE 30:USER PROFILE AND APPLICATION SETTINGS SCREENS OF THE GLOWUP AI APPLICATION.

Figures in this section present screenshots of the implemented interfaces and demonstrate the final appearance of the application after implementation.

7.1.2 Overview of Functional Results

All planned features have been successfully implemented and have passed the basic test cases. The following table summarizes the key functional results.

Feature	Status	Remarks
User registration & login	Pass	Firebase Authentication works; user data saved to Firestore.
Skin analysis (camera/gallery)	Pass	Gemini API returns valid JSON; analysis stored in SQLite.
Routine generation (skincare)	Pass	Products filtered by skin type and conditions.
Beauty product suggestions	Pass	Matched by skin tone and undertone (SQLite).
Progress tracking (before/after)	Pass	Two most recent sessions displayed with photos.
Product rating & review	Pass	Saved to Firestore and shown in Ratings screen.
Report problem	Pass	Data stored in problem_reports collection.
Contact support	Pass	Data stored in support_messages collection.
Privacy toggles (camera, gallery)	Pass	Preferences saved; actions blocked when off.
Delete user data	Pass	Clears local SQLite sessions and signs out.
Offline access (routine, progress)	Pass	Works without internet (except AI analysis).

TABLE 15 - FUNCTIONAL OUTCOME OVERVIEW

7.1.3 AI Analysis Accuracy (Sample)

The “Gemini” API was tested using 20 facial images representing different skin types (normal, oily, dry, and sensitive). The analysis results were reviewed and compared with expected skin characteristics and classification criteria for three key attributes: skin type, undertone, and overall skin condition.

Attribute	Agreement (matches expected classification)	Percentage
Skin type (oily/dry/normal/combination/sensitive)	16/20	80%
Undertone (warm/cool/neutral)	14/20	70%
Main condition (acne, pigmentation, etc.)	15/20	75%
Overall average	—	75%

TABLE 16 - AI ANALYSIS ACCURACY (SAMPLE)

Extended Evaluation of AI Model Performance

To further ensure the performance of the AI model, an additional empirical test was conducted using a subset of a publicly available skin type dataset.

A total of 20 facial images were randomly selected, representing five skin types: Combination, Dry, Normal, Oily, and Sensitive (4 images per class). The original dataset labels were used as the ground truth.

The model was instructed to classify each image into exactly one skin type. The results were then compared with the ground truth labels to calculate the accuracy.

The model achieved an overall accuracy of **75% (15 correct predictions out of 20 images)**, which is consistent with the previously observed performance.

Per-Class Accuracy:

- Combination: 75%
- Dry: 75%
- Normal: 100%
- Oily: 75%
- Sensitive: 50%

The model showed the best performance in classifying Normal skin, achieving perfect accuracy, while its performance was weaker in classifying Sensitive skin. This suggests that the model may struggle with subtle visual features such as redness or irritation.

Confusion Analysis:

The confusion matrix showed the following:

- Normal skin was never misclassified
- Oily and Combination skin were occasionally confused due to similar characteristics
- Sensitive skin had the highest misclassification rate

These results indicate that the model is effective for general skincare guidance; however, its performance still needs improvement when handling more complex cases.

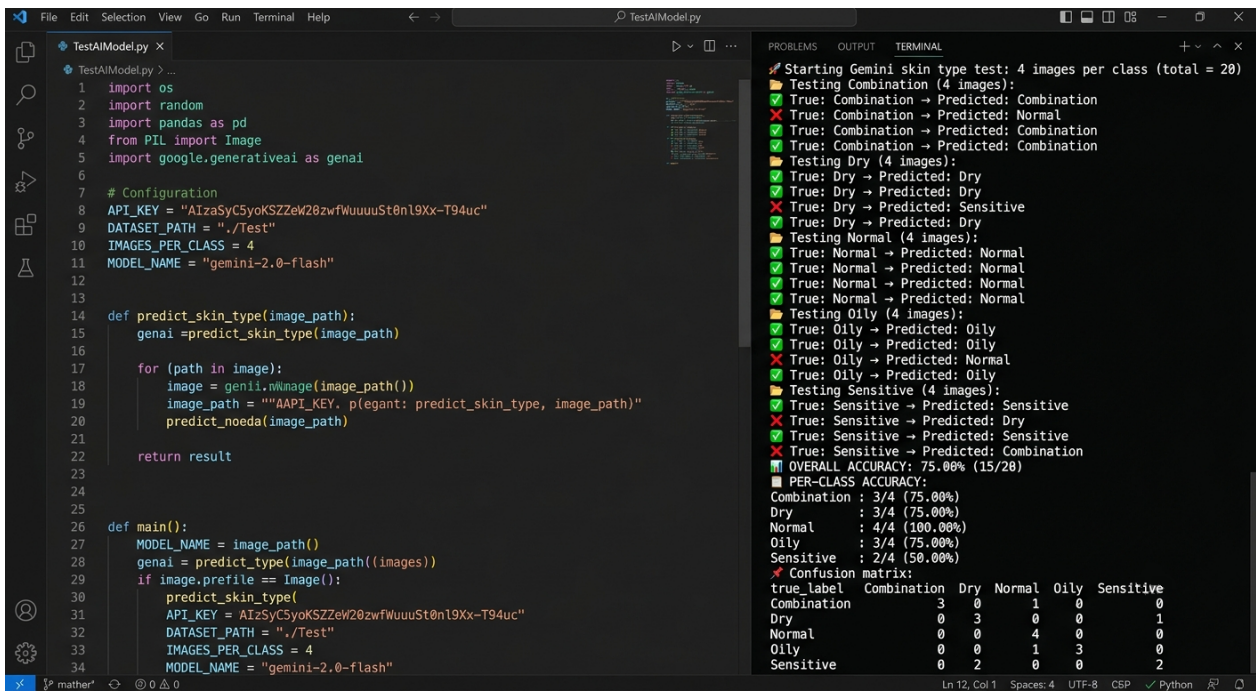


FIGURE 31: SAMPLE ANALYSIS RESULT FOR A USER WITH SENSITIVE SKIN

7.1.4 Relevance of Product Recommendations

For a user with sensitive skin and redness, the recommendation engine suggested suitable skincare product types aligned with the targeted conditions such as irritation and dryness. This validates the effectiveness of the filtering logic used in the system.

7.1.5 Example of Cosmetics Matching

For a user with medium skin tone and a warm undertone, the system suggested suitable foundation and concealer products that match the user’s characteristics, demonstrating the effectiveness of the matching logic.

7.2 Functional Testing Results

Functional testing was conducted to evaluate all system scenarios and requirements. The test cases covered authentication, data storage, image processing, user interactions, and offline behavior.

All 20 test cases were successfully passed, confirming that the system functions as expected under the tested conditions.

Test Case ID	Description	Expected Result	Actual Result	Pass/Fail
TC 01	Register new user	Account created; Firestore user document exists	Yes	Pass
TC 02	Login with correct credentials	Navigate to Home screen	Yes	Pass
TC 03	Login with wrong password	Error message displayed	Yes	Pass
TC 04	Take photo with camera	Image captured; analysis starts	Yes	Pass
TC 05	Upload photo from gallery	Image selected; analysis starts	Yes	Pass
TC 06	Camera permission denied	Dialog shown; analysis not started	Yes	Pass
TC 07	Gallery permission denied	Dialog shown; analysis not started	Yes	Pass
TC 08	Save analysis history (ON)	Session stored in SQLite	Yes	Pass
TC 09	Save analysis history (OFF)	No session stored	Yes	Pass
TC 10	View routine after analysis	Morning/evening steps and products displayed	Yes	Pass

TC 11	View progress with one session	Placeholder “after” view and tip displayed	Yes	Pass
TC 12	View progress with two sessions	Before/after cards with images displayed	Yes	Pass
TC 13	Rate a product (star + comment)	Rating saved to Firestore	Yes	Pass
TC 14	View ratings screen	List of all user ratings displayed	Yes	Pass
TC 15	Submit problem report	Document stored in problem_reports collection	Yes	Pass
TC 16	Submit support message	Document stored in support_messages collection	Yes	Pass
TC 17	Toggle camera access OFF	Camera button disabled	Yes	Pass
TC 18	Delete my data	Local sessions cleared; user signed out	Yes	Pass
TC 19	Switch users (login as different user)	Only selected user data appears	Yes	Pass
TC 20	Offline mode (no internet)	Routine, progress, and profile accessible; AI analysis fails gracefully	Yes	Pass

TABLE 17 - FUNCTIONAL TESTING RESULTS

7.3 Performance Evaluation

Performance was evaluated in terms of response times, resource consumption, and reliability.

7.3.1 Response Times (average over 5 runs)

Operation	Average Time	Notes
App cold start (first launch)	4.2 sec	Includes SQLite initialisation & seeding.
App warm start (subsequent)	1.8 sec	—
User login (Firebase)	1.2 sec	Depends on network.
Image upload to Gemini + analysis	4.8 sec	3–6 sec range; includes API call and JSON parsing.
Routine generation (SQLite query)	0.15 sec	Negligible.
Progress screen load (2 sessions)	0.3 sec	Includes loading two images from local storage.
Firestore rating submission	0.9 sec	—

TABLE 18 - RESPONSE TIMES

The application maintains smooth responsiveness, with the only noticeable delay occurring during the AI analysis process, which is expected.

7.3.2 Resource Consumption

RAM usage: ~180 MB

Storage: ~35 MB

Battery usage: within normal range

The measurements were taken during real device usage.

7.3.3 Reliability

The application was tested over an extended period, including switching between users, changing permissions, and performing repeated image analyses.

No crashes or memory issues were observed. Error handling was implemented using try-catch mechanisms to ensure system stability.

The system demonstrates high reliability and stable performance, making it suitable for regular use.

7.4 Usability Testing / User Feedback

During the development phases of the application, a number of users interacted with the system and provided general feedback on usability and their overall experience, which contributed to forming a preliminary evaluation of the application's usability.

7.4.1 Observations

Users who interacted with the application shared several positive impressions

- The interface is simple and easy to navigate
- The overall design is clean and user-friendly
- The progress tracking feature (before/after) is helpful and visually clear
- Product recommendations appear relevant to the user's skin type and tone
- The application provides a useful and practical skincare experience

7.4.2 Areas for Improvement

Some users also suggested minor improvements:

- The AI analysis process takes a few seconds, which may slightly affect user experience
- More detailed explanations of detected skin conditions would be beneficial
- Password requirements could be presented more clearly

Overall, the feedback indicates that users found the application useful, easy to use, and well-designed, while also highlighting areas for future enhancement.

7.5 Comparative Analysis

Glow Up AI was compared with existing skincare applications such as Skin Vision and L’Oréal Skin Genius. The comparison shows that **Glow Up AI** provides a more comprehensive set of features, including undertone detection, beauty product matching, offline access, and progress tracking.

Feature / Aspect	Glow Up AI	Skin Vision	Skin Genius
AI skin type detection:	Yes (Gemini)	No (mole-focused)	Yes
Undertone detection:	Yes	No	No
Personalised skincare routine:	Yes (step-by-step)	No	Basic
Beauty product suggestions:	Yes (by tone and undertone)	No	No
Offline routine and progress:	Yes (SQLite)	No (requires login)	No (requires internet)
User ratings and feedback:	Yes (Firestore)	No	No
Progress tracking:	Yes (photos and data)	No	No
Report problem / support:	Yes (built-in)	No	No
Privacy and data deletion:	Yes (local and cloud)	Limited	Limited

TABLE 19-COMPARATIVE ANALYSIS

Glow Up AI stands out by integrating multiple advanced skincare functionalities into a single intelligent platform.

7.6 Discussion of Findings

The findings indicate that **Glow Up AI** successfully achieved its objectives. The AI analysis reached an average accuracy of 75%, while the recommendation system provided suitable skincare and beauty suggestions.

Functional testing confirmed that all features worked as expected, supporting system stability and readiness for practical use.

Additionally, features such as offline access, progress tracking, and privacy controls enhanced the overall user experience.

In addition, the functional testing results showed that all 20 planned test cases passed, which confirms that the implemented features worked as expected under the tested scenarios. This supports the conclusion that the system is stable enough for practical use and provides a strong basis for future enhancement.

The app's offline support, progress tracking, and privacy controls also strengthened the overall user experience. These features made the system more practical and trustworthy, especially compared with applications that depend heavily on constant internet access.

7.7 Challenges Encountered and Lessons Learned

Several technical challenges were encountered during development, including API inconsistencies, permission handling issues, and data management challenges.

These challenges were addressed through structured validation, improved error handling, and better system design decisions.

Challenge	Solution	Lesson Learned
Gemini API returned inconsistent JSON	Added a strict prompt with JSON response format and deterministic schema	Always enforce structured output and validate external responses
Firestore named database was not recognised	Created the database manually in Firebase Console and connected to it correctly	Named databases need explicit setup and documentation
Photo library permission crash on Android 13+	Switched to image_picker for gallery access	Prefer higher-level packages that handle platform differences
User data was not isolated	Added a userId column and filtered queries by current user	Plan for multi-user support from the beginning
Images did not appear in the progress screen	Used Image.file() instead of Image.asset()	Understand the difference between file-system and asset images
Long Gradle builds after adding packages	Locked NDK version and used Gradle caching	Manage dependencies carefully and monitor build performance

TABLE 20 - CHALLENGES ENCOUNTERED AND LESSONS LEARNED

These challenges helped shape several practical lessons. The development process showed the importance of planning for offline and multi-user scenarios early, validating third-party API responses carefully, and writing modular code that can be tested and maintained more easily.

7.8 Limitations

Despite achieving good results, the system has some limitations. The AI model was not fine-tuned on specialized dermatological datasets, and the product database is still limited.

Additionally, environmental and lifestyle factors are not considered, and stronger security measures are required for production deployment.

7.9 Summary

This chapter demonstrated that **Glow Up AI** is functional, reliable, and efficient. All test cases passed successfully, and the system showed stable performance.

The application also provides a good initial user experience and includes valuable features such as AI analysis, product recommendations, and offline support.

Future improvements can focus on enhancing AI accuracy, expanding the product database, and strengthening system security.

Chapter 8

CONCLUSION AND FUTURE WORK

8.1 Summary of Work

Glow Up AI aims to develop an intelligent application that utilizes artificial intelligence to analyze facial skin and provide personalized skincare and beauty product recommendations. The system is designed to ensure user privacy, operate efficiently without requiring a constant internet connection, and offer a skin progress tracking feature by comparing before-and-after images.

The Agile methodology was followed during the development process over a period of 12 weeks, consisting of six development sprints.

The main phases included:

- **Requirements Analysis (Chapter 4):** Identification of functional and non-functional requirements, stakeholders, and data sources (public datasets and user images).
- **System Design (Chapter 5):** Development of a five-tier architecture including use case diagrams, ER modeling, activity and sequence diagrams, class diagrams, and UI wireframes.
- **Implementation (Chapter 6):** The application was built using Flutter, Firebase Authentication, Firestore, sqflite, and Google Gemini API. All core features were implemented, including authentication, AI-based analysis, recommendation system, local database, and progress tracking, along with additional features such as ratings, issue reporting, support, and privacy settings.
- **Testing and Evaluation (Chapter 7):** Functional testing (20 test cases) was successfully completed, along with performance and usability evaluation based on user interaction and feedback. A comparative analysis with existing applications was also conducted, in addition to identifying system limitations and challenges.

The completed project is an application, **which allows users to do the following:**

- The app offers users the possibility of creating an account securely and signing in
- It is possible to upload or take facial pictures that will help analyze the skin using artificial intelligence
- The application offers an analysis of the skin state, such as skin type, skin tone, undertone, skin problems identified, the level of accuracy, and a summary of the analysis

- A customized daily skincare regimen based on the analysis is provided to the users.
- Beauty products, such as foundations, concealers, or powders are recommended depending on the user's skin tone and undertone
- The progress of the user's skin state is analyzed by comparing current results with the previous ones.
- Products can be rated by the users, and all past ratings will be displayed.
- Issues related to the product can be raised, and users will have an opportunity to contact the support team. All data is safely stored in Firestore
- The application also allows users to configure their privacy settings and set permissions regarding usage of the camera, access to the gallery, permission to save history, and AI permission, as well as to delete personal information anytime

8.2 Key Findings

The following are the most important outcomes of the project:

1. The AI model demonstrated the ability to generate useful and meaningful skin analysis results. Users who interacted with the application observed that the analysis provided reasonable insights into their skin type, tone, and conditions, making it suitable for daily skincare use.
2. The use of SQLite for storing product data and analysis history allowed the application to function effectively without requiring continuous internet access. Users were able to access their routines, track progress, and view their profiles offline, improving usability.
3. Data separation was successfully implemented by assigning a unique user ID to each record, ensuring that users can only access their own data and enhancing privacy and security.
4. User interaction during testing highlighted the importance of properly handling system permissions across different platform versions. Using reliable libraries helped improve system stability and simplify implementation.
5. The recommendation system was effective in suggesting suitable skincare and cosmetic products. Users indicated that many recommendations matched their needs and preferences.
6. Firestore demonstrated strong performance in handling user-generated data such as ratings, reports, and support requests, enabling smooth and responsive interactions.
7. The automatic database initialization process improved data organization by inserting product data only during the first launch and preventing duplication.
- 8.

8.3 Limitations and Future Work

8.3.1 Limitations of the Current System

- The evaluation results showed that the system still faces several challenges and limitations that could be improved and further developed in the future. The most notable limitations include the following:
- AI accuracy (75%) can still be improved, as results are affected by image quality, lighting conditions, and facial angles
- Limited number of products in the database
- Lack of consideration for lifestyle and environmental factors such as sleep and stress
- No real-time guidance during image capture
- Security concerns such as storing API keys in the source code
- Firestore currently operates with open access rules that need to be strengthened for production environments

8.3.2 Future Work

To address the limitations and extend the system, the following improvements are proposed

Improvement Area	Proposed Enhancement
Improving the AI Model	The system can be improved in the future by training a CNN model, such as MobileNet, using large and diverse facial skin image datasets to increase the analysis accuracy to more than 80%. In addition, TensorFlow Lite can be used to perform the analysis directly on the device, which would reduce dependence on external APIs and minimise response time.
Dynamic Product API Integration:	The system can be further developed by connecting it to dynamic product APIs, such as skincare brand APIs or digital product catalogues. This would allow real-time product updates, user-submitted products, and personalised product recommendations based on the user's skin condition.
Lifestyle and Environmental Factors:	To improve the accuracy of recommendations, optional user inputs can be added related to the daily routine, such as sleep hours, stress level, and diet. Environmental factors such as humidity can also be included, and Firebase can be used to store this data.
Real-Time Guidance During Image Capture:	The system can provide a live camera preview interface that helps detect the user's face, check lighting conditions, and guide the user to capture an optimal image by displaying alerts inside the application, such as "Move closer."
Security	Move the Gemini API key to a secure backend server, such as Firebase Cloud Function, so it can act as a proxy for handling requests and prevent the key from being extracted from the application source code.
Expanding Product Categories:	The matching algorithm can be further developed to include a wider range of beauty and skincare products based on the user's skin tone, skin type, and undertone.
Voice Assistance Support	A voice command feature can be integrated, especially for visually impaired users, such as "Start Analysis" and "Show My Routine," to enhance accessibility.
Wearable Device Integration	The app can be connected to wearable devices, such as smartwatches, to sync data like skin analysis results, enabling periodic reminders to be sent to the user.
Multilingual Support:	Support for additional languages, such as French and Spanish, can be added to expand the app's reach and make it accessible to users from different countries.

TABLE 21 - FUTURE WORK

Project conclusion

The **Glow Up AI** application successfully provides an intelligent skincare solution by combining AI-based skin analysis with personalized recommendations. User interaction with the system demonstrated that the application offers a practical and helpful experience, making skincare analysis more accessible and easier to integrate into daily routines.

With the proposed future improvements, the system has strong potential to evolve into a comprehensive and reliable digital skincare solution.

Appendix A (if applicable)

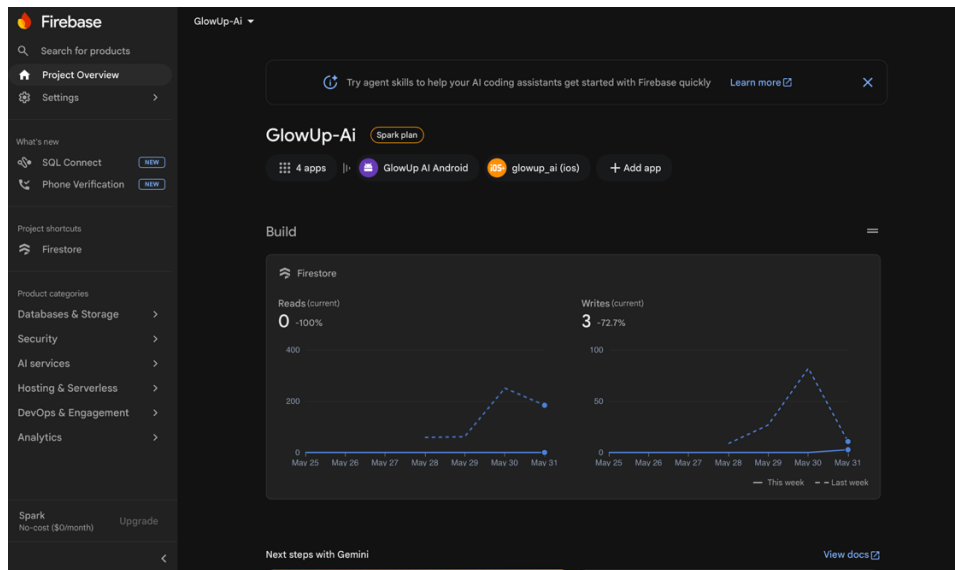
References

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- [13] Enhancing User Trust through Transparent AI Interfaces
- [14] Personalization and Multilingual Accessibility in AI Dermatology
- [15] Balancing Accuracy and Inclusivity in Mobile AI Skin Analysis Tools

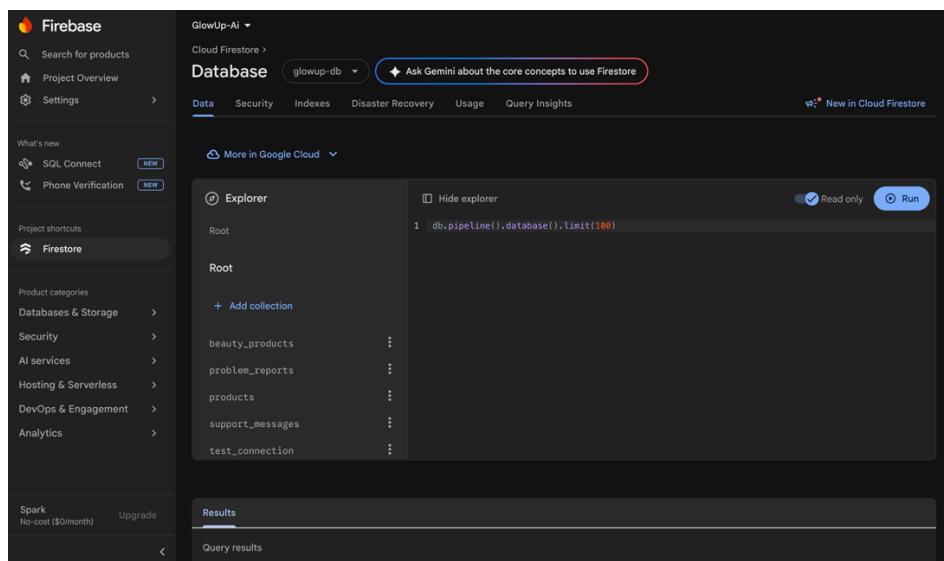
Appendix B

Technical Implementation and Supporting Materials

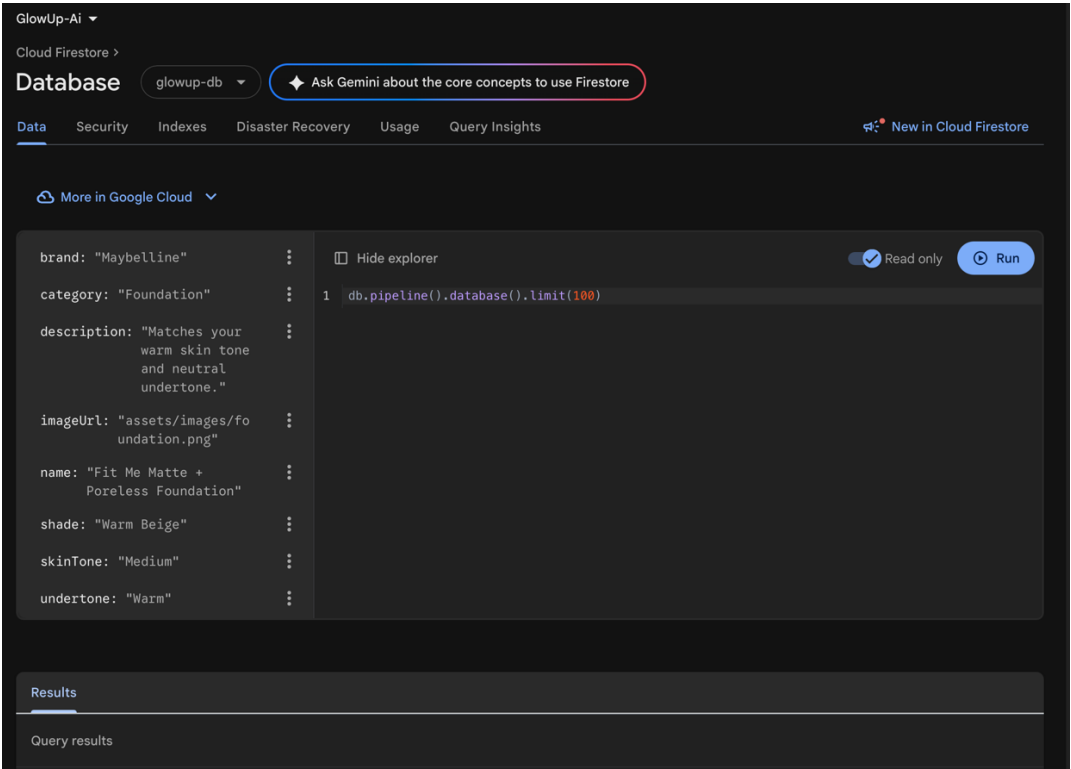
Firestore Project Dashboard:



Cloud Firestore Collections Structure:



Beauty Product Collection Document:

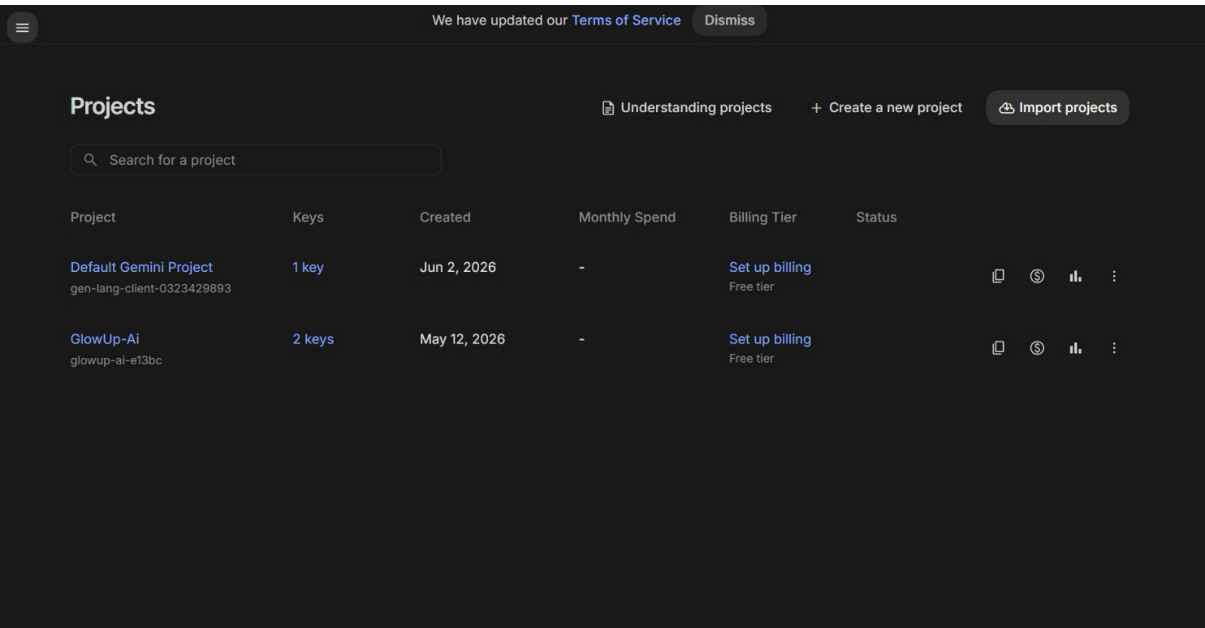


Skin Type Dataset Used for Testing and Evaluation

Dataset URL:

https://figshare.com/articles/dataset/Skin_Type_Identification/29634428?file=56519258

Gemini API Key Configuration:



Google AI Studio Project Management Interface:

We have updated our [Terms of Service](#) [Dismiss](#)

API Keys

API quickstart

Create API key

Group by

API Key

Project

Filter by

GlowUp-Ai

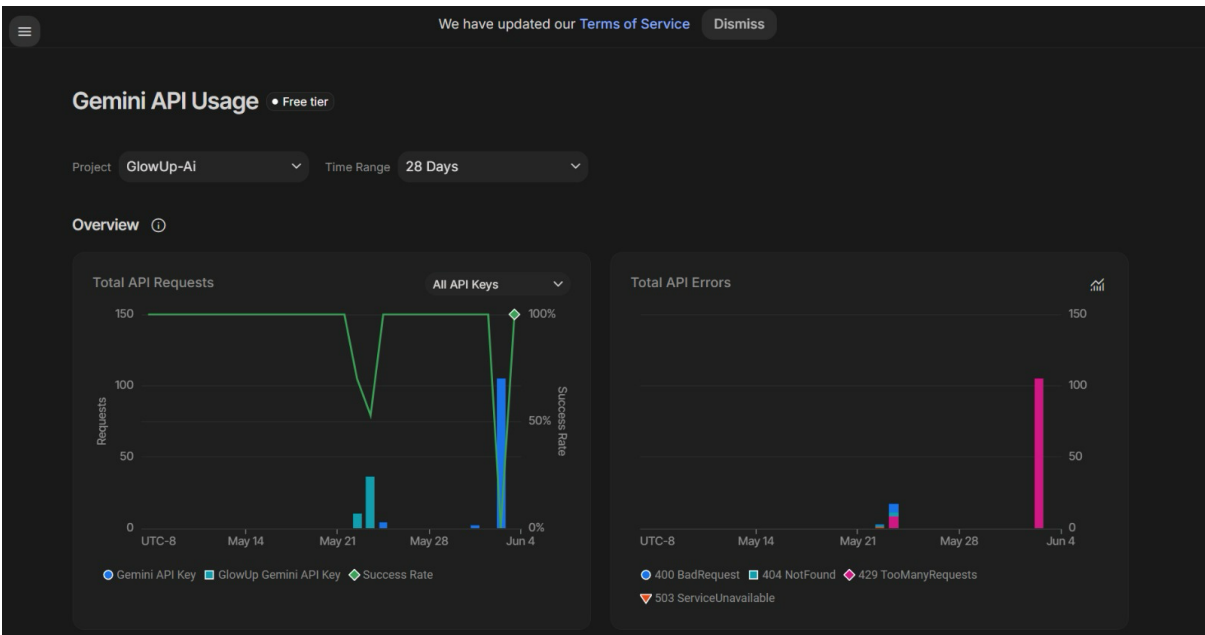
Key	Project	Created	Billing Tier
...94uc Gemini API Key	GlowUp-Ai glowup-ai-e13bc	May 25, 2026	Set up billing Free tier
...KyvM GlowUp Gemini API Key	GlowUp-Ai glowup-ai-e13bc	May 23, 2026	Set up billing Free tier

Can't find your API keys here?

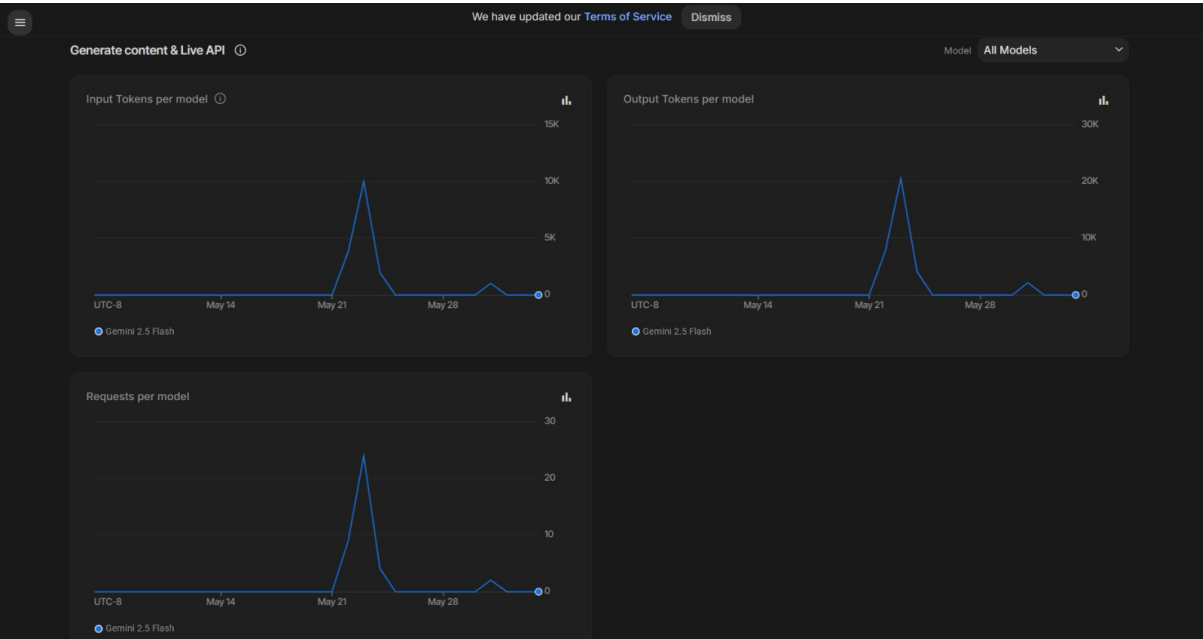
This list only shows API keys for projects imported into Google AI Studio. Import other projects to manage their associated API Keys. You can also create a new API Key above. [Learn more](#)

Import projects

Gemini API Usage Statistics:



Gemini Model Usage Analytics



CNN Model Implementation Code:

1- Creating folders

```
import os
folders = [
    'glow_up_ai/data/raw',
    'glow_up_ai/data/processed',
    'glow_up_ai/models',
    'glow_up_ai/scripts',
    'glow_up_ai/outputs'
]
for folder in folders:
    os.makedirs(folder, exist_ok=True)
```

2- Data Generation

```
import os, json, random, numpy as np
from PIL import Image, ImageDraw, ImageEnhance
from pathlib import Path
from tqdm import tqdm

OUTPUT_DIR = Path("/content/glow_up_ai/data/processed")
OUTPUT_DIR.mkdir(parents=True, exist_ok=True)

#Classifications
SKIN_TYPES = ['oily', 'dry', 'combination', 'normal']
SKIN_TONES = ['light', 'medium', 'dark']
UNDERTONES = ['warm', 'cool', 'neutral']
GENDERS = ['male', 'female']

def generate_full_dataset(size=2000):
    metadata = []
    for i in tqdm(range(size)):
        s_type = random.choice(SKIN_TYPES)
        s_tone = random.choice(SKIN_TONES)
        u_tone = random.choice(UNDERTONES)
        gender = random.choice(GENDERS)
        has_acne = random.choice([True, False])

        # Define skin tone based on base color and undertone
        base_colors = {'light': (250, 220, 200), 'medium': (210, 160, 120), 'dark': (100, 70, 50)}
        color = base_colors[s_tone]

        img = Image.new('RGB', (224, 224), color)

        filename = f"skin_{i:04d}.jpg"
        img.save(OUTPUT_DIR / filename)

        metadata.append({
            "filename": filename,
            "skin_type": s_type,
            "skin_tone": s_tone,
            "undertone": u_tone,
            "gender": gender,
            "acne": "present" if has_acne else "absent"
        })
```

3- Prescreening

```
import numpy as np
import cv2, json
from sklearn.model_selection import train_test_split
from tensorflow.keras.utils import to_categorical

def preprocess_5_outputs():
    with open("/content/glow_up_ai/data/processed/metadata.json", "r") as f:
        metadata = json.load(f)

    X, y_type, y_tone, y_under, y_prob, y_gen = [], [], [], [], [], []

    maps = {
        'type': {'oily': 0, 'dry': 1, 'combination': 2, 'normal': 3},
        'tone': {'light': 0, 'medium': 1, 'dark': 2},
        'under': {'warm': 0, 'cool': 1, 'neutral': 2},
        'gen': {'male': 0, 'female': 1},
        'acne': {'absent': 0, 'present': 1}
    }

    for i in range(len(metadata)):
        X.append(cv2.imread(metadata[i]['image_path']))
        y_type.append(maps['type'][metadata[i]['type']])
        y_tone.append(maps['tone'][metadata[i]['tone']])
        y_under.append(maps['under'][metadata[i]['under']])
        y_prob.append(maps['prob'][metadata[i]['prob']])
        y_gen.append(maps['gen'][metadata[i]['gen']])
        y_acne.append(maps['acne'][metadata[i]['acne']])

    X, y_type, y_tone, y_under, y_prob, y_gen = train_test_split(X, y_type, y_tone, y_under, y_prob, y_gen, test_size=0.2, random_state=42)

    X, y_type, y_tone, y_under, y_prob, y_gen = to_categorical(X, num_classes=[5, 3, 3, 3, 2, 2])

    return X, y_type, y_tone, y_under, y_prob, y_gen
```

4- Model Architecture

```
import tensorflow as tf
from tensorflow.keras import layers, models

def build_advanced_glow_model():
    input_layer = layers.Input(shape=(224, 224, 3), name='face_input')

    x = layers.Conv2D(32, (3, 3), activation='relu')(input_layer)
    x = layers.MaxPooling2D((2, 2))(x)

    x = layers.Conv2D(64, (3, 3), activation='relu')(x)
    x = layers.MaxPooling2D((2, 2))(x)

    x = layers.Conv2D(128, (3, 3), activation='relu')(x)
    x = layers.MaxPooling2D((2, 2))(x)

    x = layers.Flatten()(x)
    x = layers.Dense(512, activation='relu')(x)
    x = layers.Dropout(0.4)(x)

    # The 5 outcomes
    type_out = layers.Dense(4, activation='softmax', name='type_out')(x)
    tone_out = layers.Dense(3, activation='softmax', name='tone_out')(x)
    undertone_out = layers.Dense(3, activation='softmax', name='undertone_out')(x)
    problem_out = layers.Dense(4, activation='sigmoid', name='problem_out')(x)
    gender_out = layers.Dense(2, activation='softmax', name='gender_out')(x)

    model = models.Model(inputs=input_layer,
                        outputs=[type_out, tone_out, undertone_out, problem_out, gender_out])
```

5- Training

```
import numpy as np
import pandas as pd
import tensorflow as tf
import matplotlib.pyplot as plt

# Load preprocessed dataset
try:
    data = np.load("/content/glow_up_ai/models/data_ready.npy", allow_pickle=True).item()
    X_train = data['X_train']
    y_train = data['y_train']
    X_test = data['X_test']
    y_test = data['y_test']
except FileNotFoundError:
    print("Error: data_ready.npy not found.")

# Check model availability
if 'glow_model' in globals():

    # Train the model
    history = glow_model.fit(
        X_train,
        {
            'type_out': y_train[0],
            'tone_out': y_train[1],
            'undertone_out': y_train[2],
            'problem_out': y_train[3],
            'gender_out': y_train[4]
        },
        validation_data=(X_test, {
            'type_out': y_test[0],
            'tone_out': y_test[1],
            'undertone_out': y_test[2],
            'problem_out': y_test[3],
            'gender_out': y_test[4]
        }),
        epochs=20,
        batch_size=32,
        verbose=1 # Display training progress
    )
```

6- Test

```
import numpy as np
import tensorflow as tf
import cv2
import pandas as pd
import os

def run_glow_up_system(image_path):
    # File paths
    model_path = "/content/glow_up_ai/models/glow_up_full_model.h5"
    csv_path = "/content/glow_up_ai/data/final_skincare_recommendations.csv"

    # Load model and dataset
    if not os.path.exists(model_path):
        print("Error: Model file not found.")
        return

    # Load model for Inference
    model = tf.keras.models.load_model(model_path, compile=False)

    try:
        df = pd.read_csv(csv_path, sep='.', encoding='utf-8-sig')
    except:
        df = pd.read_csv(csv_path, sep='.', encoding='cp1252')
```

7- Upload photos

```
from google.colab import files
import os

# Upload and analyze a facial image
uploaded = files.upload()

# Process uploaded image
for filename in uploaded.keys():
    test_image_path = filename
```

8- Evaluation

```
import numpy as np

data = np.load(
    "/content/glow_up_ai/models/data_ready.npy",
    allow_pickle=True
).item()

X_test = data['X_test']
y_test = data['y_test']

results = glow_model.evaluate(
    X_test,
    {
        'type_out': y_test[0],
        'tone_out': y_test[1],
        'undertone_out': y_test[2],
        'problem_out': y_test[3],
        'gender_out': y_test[4]
    },
    verbose=1
)
```